Biome

USER GUIDE · v2.5 · October 2026 edition

Biodiversity Net Gain habitat mapping for QGIS
Baseline & proposed habitats · condition assessment · desk study
field survey · dashboard · maps · statutory metric export

1 Save project
2 Draw red line
3 Map baseline
4 Design proposed
5 Desk study
6 Check & export

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For QGIS 3.22+ and QGIS 4 · England · Statutory Biodiversity Metric workflow

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1 Biome and BNG in context

Biome is a QGIS dock-panel plugin that takes you through a full Biodiversity Net Gain assessment — from the red line boundary, through baseline and proposed habitat mapping, condition assessment and a live desk study, to finished maps and a pre-filled statutory metric spreadsheet — without leaving QGIS.

1.1 What BNG requires

BNG is a legal requirement under the Environment Act 2021, which inserts Schedule 7A into the Town and Country Planning Act 1990. Unless the development is out of scope or exempt, planning permission in England carries a condition requiring at least a 10% increase in biodiversity value over the pre-development baseline, secured and managed for at least 30 years. Biodiversity value is measured with the government's Statutory Biodiversity Metric, and habitats are classified using the UK Habitat Classification (UKHab).

The duty has been phased in: mandatory for major developments from 12 February 2024, for small sites from 2 April 2024, and for Nationally Significant Infrastructure Projects whose development consent applications are made from 2 November 2026. Where it applies, the applicant submits a Biodiversity Gain Plan showing the metric calculations, and a Habitat Management & Monitoring Plan where required for the 30-year period.

1.2 Changes effective from 6 August 2026

Change	Current position
Development no larger than 0.2 ha	A planning application made on or after 6 August 2026 is exempt where the red-line site area is 0.2 ha or below, unless the development impacts on-site priority habitat.
Temporary development	Exempt where the whole development is temporary and permission is granted for five years or less, unless on-site priority habitat is impacted.
Self-build and custom build	The former exemption does not apply to new applications made on or after 6 August 2026. Transitional protection applies to qualifying earlier applications and variations.
Minor-development gain hierarchy	A qualifying minor development that is not exempt may consider on-site and registered off-site delivery together and may choose off-site delivery without first exhausting on-site options. Statutory credits remain the last resort.
De minimis	The separate thresholds remain less than 25 m2 of on-site area habitat and less than 5 m of on-site linear habitat. The exemption is unavailable where on-site priority habitat is impacted.

NOTE

The 0.2 ha exemption is based on the whole site area within the planning red line, not the area of habitat loss. It is separate from the de minimis exemption, which is based on habitat impact.

THE RULE

The figures change with policy. Metric amendments and new exemptions are consulted on regularly; always confirm the current rules on GOV.UK before relying on a threshold for a submission.

1.3 The metric's three modules

The metric scores three habitat modules separately, and each must reach the 10% gain on its own — surplus in one can never make up a shortfall in another:

Module	Measured in	Biome layers
Area habitats	hectares	Baseline / Proposed Habitats (polygons) — individual trees fold into this module too
Hedgerows	kilometres	Baseline / Proposed Hedgerows (lines)
Watercourses	kilometres	Baseline / Proposed Watercourses (lines)

NOTE

Individual trees are part of the AREA module, not a module of their own — the metric converts each tree to an area-habitat equivalent by size class. Biome maps trees with a separate tool and layer purely for ease of use; when it totals units and writes the Filled Metric, trees fold into the area figures.

1.4 How Biome guards against common BNG mistakes

Most BNG assessments fail review for a small set of recurring reasons. Biome is built to surface those errors while you still have time to fix them. This table is the spine of the guide — each safeguard links to the chapter that explains it.

Common mistake	How Biome helps	See
Gaps — land inside the red line left unclassified	Dashboard coverage check flags unmapped ground	5, 13
Overlapping polygons double-count area	Overlaps detector; avoid-overlap drawing; cookie-cutter placement	5
Trading units across modules	Modules kept strictly separate in the Dashboard, classifier and Enhance recommender	13, 14
Forgetting that losses come into the 10%	The net-difference classifier subtracts the full baseline of any parcel that changes type	7

Removed-and-replaced hedges scored as retained	Pair-linked hedgerow/watercourse twins record the true fate — including the temporal discount on replacements	8
Optimistic condition with no evidence	Criterion-by-criterion condition sheets, field photos, exported assessment sheets	6, 11
Missed constraints and irreplaceable habitats	Live desk study with reports and print-ready maps	12
Planar-measure errors inflating areas ~2.6x	Every figure is measured ellipsoidally on the ground, never in projected Web-Mercator units	16

2 Installing and first run

One plugin serves QGIS 3.22+ and QGIS 4 — same folder, same data, both interfaces.

2.1 Requirements

- QGIS 3.22 or later, or QGIS 4 — Biome is fully dual-version and follows the OS light/dark colour scheme automatically on QGIS 4.
- An internet connection for basemaps, geocoding and the live desk-study constraint services. An optional offline Ordnance Survey basemap is bundled (see 4.4).
- A saved QGIS project (.qgs or .qgz) — Biome stores its data in a GeoPackage beside it and prompts you if the project is unsaved.

2.2 Install and update

Copy the biome folder into your QGIS plugins directory, then enable it in Plugins > Manage and Install Plugins. A toolbar icon and a Plugins > Biome menu entry appear; the dock opens on the right.

OS	Plugins directory
Windows	C:\Users\<you>\AppData\Roaming\QGIS\QGIS3\profiles\default\python\plugins\
macOS	~/Library/Application Support/QGIS/QGIS3/profiles/default/python/plugins/
Linux	~/.local/share/QGIS/QGIS3/profiles/default/python/plugins/

To update, replace the folder with the new version and toggle the plugin off and on in the Installed tab (or restart QGIS). Your GeoPackage data is never touched by a plugin update.

2.3 First run

The first time the panel opens, a short welcome wizard walks through the six core steps — save the project, draw the red line, map baseline habitats, design the proposed scheme, run the desk study, then check and export. Reopen it any time with Getting Started; open this guide with the User Guide button beside it.

2.4 Light and dark mode

Every colour in the panel comes from a live theme. On QGIS 4 the plugin follows the operating system's light/dark scheme and switches instantly when it changes; on QGIS 3 it matches the QGIS application style. Biome Mode (in the panel settings) optionally restyles the whole QGIS window to match the plugin.

2.5 API keys and external accounts

Biome works without any paid API account. Most mapping and desk-study sources are open services and require no key. Optional keys unlock OS Outdoor mapping, the OS Premium basemap and detailed NGD water polygons, Google's 3D Spotlight export, and Scottish TPO data. Enter keys through Settings > Connections. Keys are stored in the local QGIS user profile; never put an API secret into Biome.

Biome setting	Service to enable	Used for
Ordnance Survey API key	OS Maps API	OS Outdoor raster basemap
OS NGD premium key	OS NGD API - Features and OS NGD API - Tiles	OS Premium basemap and detailed standing-water polygons
Google Maps API key	Google Map Tiles API	Photorealistic 3D Spotlight export
Spatial Hub Scotland key	Spatial Hub user authentication key	Scottish Tree Preservation Order searches

Ordnance Survey Outdoor key

Open osdatahub.os.uk and create or sign in to an OS Data Hub account. The OpenData Plan is sufficient. Select API Projects in the left-hand menu, choose Create a new project, enter a project name such as Biome, and create it. Open the new project and select Add an API to this project. Choose OS Maps API from the catalogue and add it. The required credentials are then shown in the project. Copy the Project API Key and, in Biome, open Settings > Connections > Ordnance Survey API key, paste the key, and select Validate & Save. Do not copy the Project API Secret.

Ordnance Survey Premium NGD key - no payment card required

> **THE RULE**
>
> You do not need to enter credit-card or debit-card details to obtain or use the OS Premium Plan's monthly free allowance. Leave the Data Hub account without a payment card. If the free premium allowance is used up, OS pauses premium access until the allowance resets at the start of the next calendar month; it cannot charge a card that has not been added.

At osdatahub.os.uk, sign in and open Plans. Apply for the Premium Plan and complete the organisation and intended-use details accurately. The Premium Plan is intended for eligible OS Partners building an application or service for third parties. Eligible public-sector users may instead use the Public Sector Plan. When payment details are offered, skip that step and do not add a card.

After the plan is active, select API Projects in the left-hand menu and choose Create a new project. Enter a name such as Biome Premium, create the project, and choose Live mode for real consultancy work. Development mode is only for development and must not be used for live client work. Open the new project and select Add an API to this project. Choose OS NGD API - Features and OS NGD API - Tiles from the available APIs and add both to the project.

Open the completed project and copy its Project API Key. Both NGD APIs use that one project key. In Biome, open Settings > Connections > OS NGD premium key, paste it and select Validate & Save. Do not enter the Project API Secret. The Premium basemap is then available from the basemap bar, and the standing-water search automatically adds NGD polygons while retaining its free fallback sources.

> NOTE
>
> To confirm the no-card safeguard, open the OS Data Hub account's Billing & Payments page and check that no payment card is registered. The free allowance is shared across premium OS APIs. Data Hub provides usage information and threshold notifications.

Google Maps key for 3D Spotlight

Open console.cloud.google.com, create or select a Google Cloud project, attach a billing account, then open APIs & Services > Library. Search for Map Tiles API and select Enable. Open APIs & Services > Credentials, choose Create credentials > API key, copy the key, and restrict it to the Map Tiles API where practical. In Biome, open Settings > Connections > Google Maps API key, paste it and select Validate & Save.

> COMMON BNG PITFALL
>
> Google's Map Tiles API is separate from the no-card OS arrangement. Google requires billing to be enabled. Review Google's current pricing and set appropriate quotas or budget alerts before using the 3D export.

Spatial Hub Scotland authentication key

Open data.spatialhub.scot, select Register, and create a free user account. A unique authentication key is issued automatically with the account and is shown on the user's profile and web-service information. Copy the authentication key only - not a complete WFS address. In Biome, open Settings > Connections > Spatial Hub Scotland key, paste it and select Validate & Save. Open datasets are available with a free account; datasets marked premium require the appropriate PSGA or commercial access.

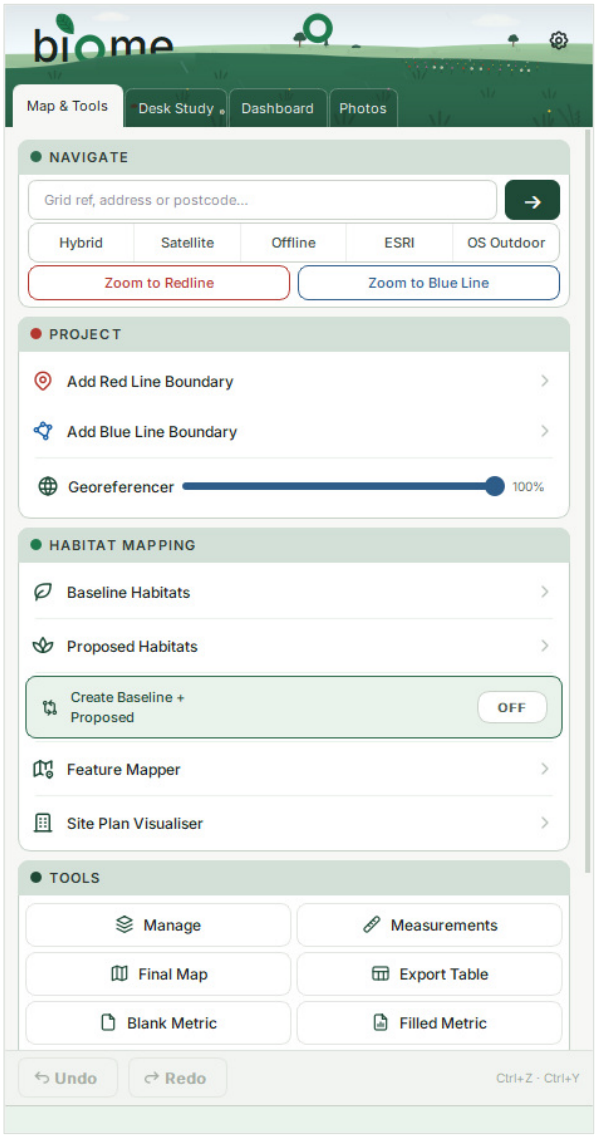
If no key is entered

Biome continues to operate. Hybrid, Satellite, ESRI and Offline basemaps remain available; standing-water searches continue with the free OS Open Zoomstack, OpenStreetMap, Priority Habitat Inventory and HabMoS sources. Only the optional service associated with the missing key is unavailable.

3 The panel at a glance

Biome is a single dock panel with five tabs. Map & Tools is home; the others are review and reference surfaces.

Tab	Purpose
Map & Tools	The workflow home — Navigate, Project, Habitat Mapping and Tools cards.
Desk Study	Live constraint data and pre-survey mapping tools (chapter 12).
Dashboard	Live unit balance, obligations, trading slots, credits and data-quality checks (chapter 13).
Photos	Field-survey features that carry photos (chapter 11).
Enhance	On-site shortfall solver and off-site routes (chapter 14).



The Map & Tools home screen in QGIS, showing the main navigation, project, habitat-mapping and export controls.

3.1 The Map & Tools tab cards

- Navigate — combined address / postcode / grid-reference search, Zoom to Redline and Zoom to Blue Line, and the basemap selector.
- Project — Add Red Line Boundary, Add Blue Line Boundary and the Georeferencer, used most often to align architects' plans with the mapped site.
- Habitat Mapping — Baseline Habitats, Proposed Habitats, the Create Baseline + Proposed toggle, Feature Mapper and Building & Tree Mapper.
- Tools — Manage, Overlaps, Measurements, Final Map, Export Table, the metric exports and Import Field Survey.

A hint bar at the foot of the panel describes whatever button you hover over, in plain English.

3.2 Undo and redo — everywhere

The Undo button (with its counter) unwinds the most recent placement wherever it was made — a habitat parcel, a tree, an ecological feature, a duplicate, a hedge trim — and Redo brings it back. Each entry removes everything the action placed (including features split across layers) and restores anything it modified. The stack is saved with the project, so undo still works after closing and reopening — and every entry is identity-checked, so it can never delete a feature that merely inherited a recycled internal id.

3.3 The settings cog

All per-feature settings live behind the one cog on the main panel: desk-study search radii, map titles & headings, date format, live layer refresh, measurement preferences and more. Each row shows its current status, so you can see at a glance what is set.

3.4 The floating layer panel

In Biome Mode the native QGIS Layers panel is replaced by a compact floating overlay on the canvas listing the Biome layers with visibility toggles — the full layer tree stays available in the normal QGIS panel whenever you need it.

4 Setting up a project

Save first, draw the red line deliberately, and pick the right basemap — everything downstream hangs off these steps.

4.1 Save the project first

Before drawing anything, use Project > Save As and save as either .qgs or .qgz. Biome creates a GeoPackage named after the project (for example, MySite_bng_data.gpkg) in the same folder and writes every feature there as you go. The plugin will not place features until the project is saved. Keep one folder — one QGIS project file and one .gpkg — per site.

4.2 Finding the site

The Navigate search accepts a postcode, an address or an OS grid reference and zooms straight there. Address search uses OpenStreetMap's Photon geocoder with a postcode-centroid fallback, so house numbers that aren't individually mapped still land in the right street.

4.3 Red line and blue line

Click Add Red Line Boundary and digitise the application site: left-click to place vertices, right-click to finish. The red line defines the area over which losses and gains are measured, and drives the site area, centroid, clip-to-boundary and every desk-study query — it is the single most consequential boundary in the assessment.

> **COMMON BNG PITFALL**
>
> Drawing the red line too loosely — or around land you do not control — changes the baseline the whole metric is built on. Digitise it deliberately and check it against the application drawing before mapping habitats.

Add Blue Line Boundary draws the wider ownership boundary. Use it when the applicant controls land beyond the application site — the Enhance tab's off-site route can score blue-line land under the off-site rules (chapter 14).

4.4 Basemaps

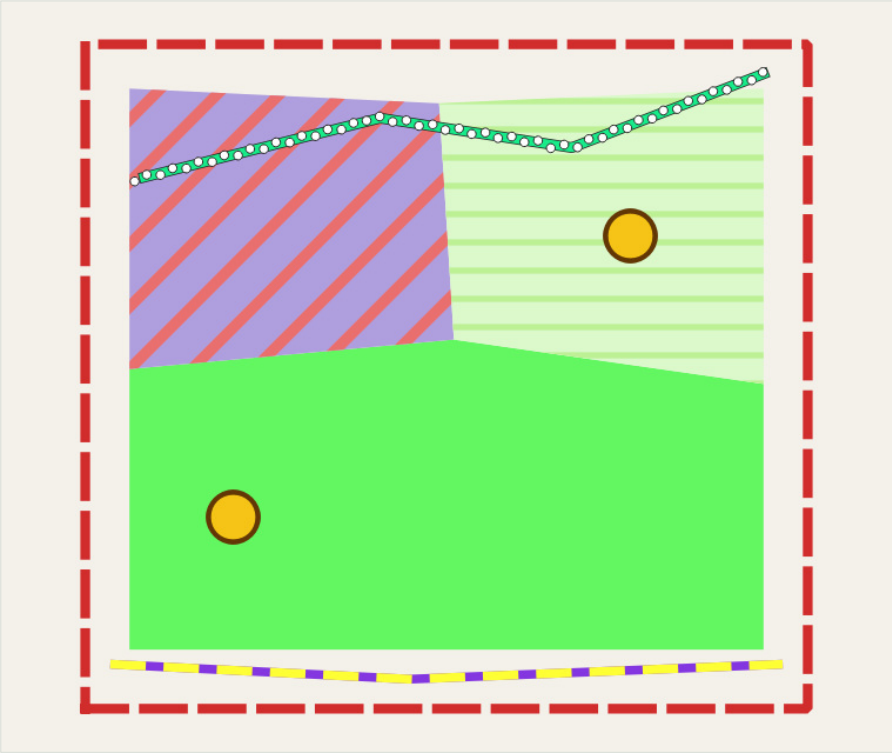
The basemap selector switches between aerial imagery, streets, OS Outdoor, the optional OS Premium vector basemap and the bundled offline OS Zoomstack basemap, which works with no connection and no API key. Right-click the Offline button (or press it a second time) for its feature toggles — roads, airports, labels and other layer groups can be switched off individually and the choice is remembered. Basemaps are session-only: they are stripped from the saved project so a project opened elsewhere never complains about missing layers. The Premium button's arrow opens Road, Outdoor, Light and Black and White styles without activating the map. Choose which basemap buttons are shown under Settings > Map & Tools > Basemap buttons.

4.5 Georeferencing architects' plans

The Georeferencer is primarily intended for plans supplied by architects. Load the plan as an image, match recognisable points on the drawing to the same positions on the map and run the transformation. The aligned plan is then available as a raster layer beneath or alongside the red line, habitat mapping and ecological features. This makes it easier to compare the proposed layout with the surveyed site, trace relevant development footprints and include the plan in a Final Map. The same workflow can also be used for drone orthophotos and other suitable imagery.

5 Mapping the baseline

The baseline is the highest-stakes step in any BNG assessment: every unit of gain is measured against it, so an inaccurate or incomplete baseline undermines everything downstream.



A baseline in progress, rendered with Biome's real symbology — area parcels, a hedgerow (bottom), a watercourse (top) and individual trees inside the red line.

5.1 The habitat form

Open Baseline Habitats and choose what to map — Area, Hedgerow, Watercourse or Trees. A global habitat search sits above the type lists, so you can jump straight to any of the 157 habitat types by name. The form captures the fields the statutory metric needs:

Field	Notes
Broad habitat & sub-type	The UKHab classification (e.g. Grassland > Modified grassland). The sub-type drives the distinctiveness band.
Condition	Good to Poor — assessed against the metric's condition criteria, ideally through the built-in condition sheets (chapter 6).
Strategic significance	Whether the parcel sits within a Local Nature Recovery Strategy area — applies the 1.0 / 1.1 / 1.15 multiplier.

Habitat ref & comments	Free-text cross-references to your survey, carried through to the exported table.

> **NOTE**
>
> You no longer enter a retention category by hand. Biome derives each parcel's fate — retained, enhanced, lost, created — from the mapping itself (chapter 7), which removes a whole class of bookkeeping errors.

5.2 Drawing area habitats

- Standard mode — click each vertex; right-click to finish. A live counter beside the cursor shows the area as you draw.
- Freehand mode — hold and drag to trace organic shapes such as ponds or woodland edges; release to close, Escape to cancel.
- Clip to Redline / Blue Line — anything drawn outside the chosen boundary is trimmed off on save, so you need not align edges precisely.
- Continuous placement — the form stays armed after each feature, so you can sweep a whole field pattern without re-opening it. Pressing any other panel button disarms the tool safely.

5.3 No gaps, no overlaps

Every piece of land inside the red line should be classified, even amenity grass, hardstanding or buildings. Unmapped gaps quietly understate the baseline and inflate the apparent gain — a frequent reason assessments are queried at review.

Three drawing aids keep the coverage clean: snapping and avoid-overlap are enabled automatically while you draw, so polygons butt tightly against their neighbours; the cookie cutter option punches a new parcel's footprint OUT of any parcel beneath it, so mapping a pond inside a meadow never double-counts the water; and the Overlaps tool scans every habitat layer for residual intersections and zooms you to each conflict.

5.4 Hedgerows, watercourses and trees

Hedgerows (13 types) and watercourses (5 types) are drawn as lines with the same live length counter; individual trees are placed as points with a size class (Small to Very Large, urban or rural). Chapter 8 covers how their baseline/proposed pairing works; chapter 9 covers trees in detail, including CSV import from an arboricultural schedule.

6 Assessing habitat condition

Condition is a major driver of unit value — and the part of an assessment most often challenged. Biome builds the evidence in as you map.

6.1 Condition sheets

Every assessable habitat class has a built-in condition sheet reproducing the statutory metric's criteria. Tick the criteria that pass, and the sheet computes the condition grade for you — including the special counting rules (essential criteria, n-of-m thresholds and the woodland scoring system). The completed sheet is stored on the feature itself, so reopening the assessment months later shows exactly which criteria you ticked.

> **HOW BIOME HELPS**
>
> A grade backed by a criterion-by-criterion sheet survives review; a bare “Good” does not. The sheets can be exported in the statutory format (chapter 18) and paired with geotagged photos from the Field Mapper (chapter 11).

6.2 Where to assess

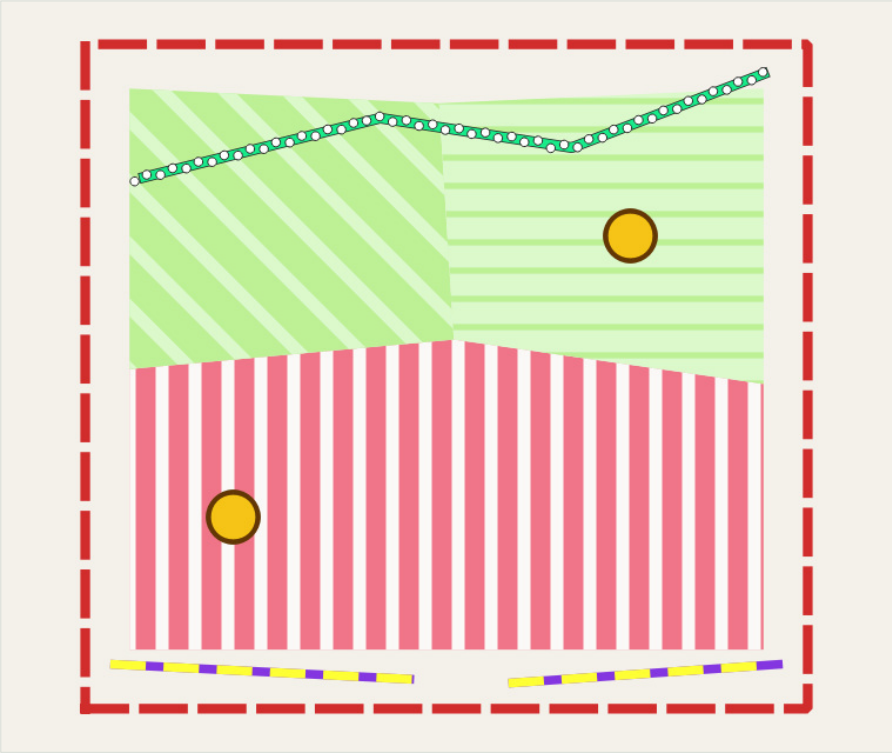
- While placing — the habitat form offers the sheet as you map each parcel.
- From Manage — every assessable feature row carries a clipboard button to open, copy or export its sheet later.
- In the field — the Mobile Field Mapper carries the same sheets, computed with identical logic, so a condition assessed on the phone imports with its full detail.

6.3 Condition in linked pairs

For a retained hedgerow, watercourse or tree (a linked baseline/proposed pair, chapter 8), edit the proposed twin's condition to record what the scheme does to it: a better condition scores as enhancement; a worse one records the honest outcome — the baseline value is lost and the proposed feature counts as creation at its lower condition. The metric has no “degrade in place” pathway, and Biome follows the statutory treatment automatically.

7 Designing the proposed scheme

Proposed habitats record the post-development design. Biome derives every parcel's fate from the mapping, so the metric always reflects what is actually drawn.



The same site as chapter 5, redesigned: sealed development replaces the lower grassland, a meadow is created top-left, and the retained hedgerow has been trimmed where the access road crosses it.

7.1 Losses come into the 10%

When a parcel changes habitat type, its full baseline value is lost and must be made good before any gain counts. Biome runs all calculations through a single net-difference classifier shared by the Dashboard and the metric export: for each habitat and condition it matches baseline against proposed, scores the overlap as retained, pairs valid condition uplifts as enhancement, and counts everything else as loss and creation. The Dashboard can never disagree with the export because they read the same engine.

7.2 Create Baseline + Proposed

Turn the Create Baseline + Proposed toggle on before drawing and each baseline feature you place also creates a matching proposed twin in the same position — the natural starting point where much of the site is retained. Edit or delete the twin afterwards to record change: for area habitats the classifier scores the geometric difference; for hedgerows, watercourses and trees the twins are pair-linked (chapter 8).

7.3 Enhancement, properly

The metric only accepts certain changes as enhancement — same broad group, achievable condition uplifts, permitted habitat changes. Biome's enhancement engine mirrors those rules: map-drawn enhancements fill the metric's enhancement sheet with the correct baseline linkage, and disallowed combinations are scored as loss + creation rather than silently mis-labelled as enhancement.

7.4 Trading rules in practice

Replacement must be like-for-like or like-for-better within the same module: no trading down in distinctiveness, Medium losses replaced within the same broad group, High and Very High habitats needing bespoke treatment. The Dashboard's trading slots (chapter 13) show each obligation and whether the current design satisfies it.

8 Hedgerows, watercourses and the trimmer

Linear habitats are scored feature-by-feature through linked baseline/proposed pairs — including the everyday case of a hedge cut by a new access road.

8.1 Why pairs?

For area habitats, netting quantities is correct practice. For an individual hedge or tree it is not: removing a hedge and planting an equal length elsewhere is NOT retention — the replacement carries a temporal discount while it grows. Biome therefore links each baseline hedgerow, watercourse and tree to its proposed twin with a hidden pair id:

Pair state	Classified as
Baseline + proposed twins linked, same type & condition	Retained
Linked, proposed twin improved (condition or eligible type)	Enhanced
Linked, proposed twin downgraded	Lost + Created at the lower state (the statutory treatment)
Baseline feature with no twin	Lost
Proposed feature with no twin	Created

Placing with the paired toggle (7.2) links twins automatically. A feature placed standalone can be linked afterwards with the Manage tab's duplicate button, which copies it to the other phase as a retained twin.

8.2 The hedge trimmer

> **NEW**
>
> The trimmer is new in this edition: cut a section out of any hedgerow or watercourse — for an access road, a visibility splay, a culvert — directly from its Manage row.

Every hedgerow and watercourse row in Manage Features has a scissors button. Click it, then click two points on the highlighted line: a red preview and a floating counter show the section being removed; confirm and it is cut out. The tool stays armed for further cuts; right-click or Escape finishes.

The trim is directional — only the layer you armed it on changes visibly, and the metric records the honest fate underneath:

- Trim a proposed hedge and the gap appears in the proposed scheme; the baseline hedge underneath stays exactly as mapped, and the trimmed length counts as Lost.

- Trim a baseline hedge and the reverse happens: the proposed line is untouched and the extra length counts as Created.
- Kept sections stay properly paired on both sides, so retained lengths keep scoring as retained — and one press of Undo restores the whole trim exactly.

8.3 Watercourse extras

Watercourses carry the metric's extra fields — encroachment and left/right riparian encroachment — and the classifier recognises delivery improvements (for example reduced encroachment) as enhancement even where type and condition are unchanged, exactly as the statutory watercourse module does.

9 Individual trees

Trees are mapped as points, sized by class, and scored inside the area module — with imports for arboricultural schedules.

9.1 Placing and pairing

Place baseline trees by size class (Small, Medium, Large, Very Large; urban or rural) — each converts to the statutory area-equivalent for its class. With the paired toggle on, each tree gets a retained proposed twin; deleting the twin marks the tree lost, and newly planted proposed trees render green on the map with their own “Created” legend entries.

Place a tree near an unlinked feature in the opposite phase and Biome asks whether it is the same physical tree (retained) or a new/replacement planting — the honest answer drives the temporal discount.

9.2 CSV import

The Site Plan Visualiser card imports a tree schedule from CSV — coordinates are auto-detected (OSGB eastings/northings or lat/long), a preview shows what will land where, and each tree arrives with its size class ready for the metric.

> NOTE
>
> A BS5837 category grades safety and amenity; BNG condition grades ecological value. They are different scales — a category-U tree full of deadwood can still score well for condition criteria that reward decay habitat.

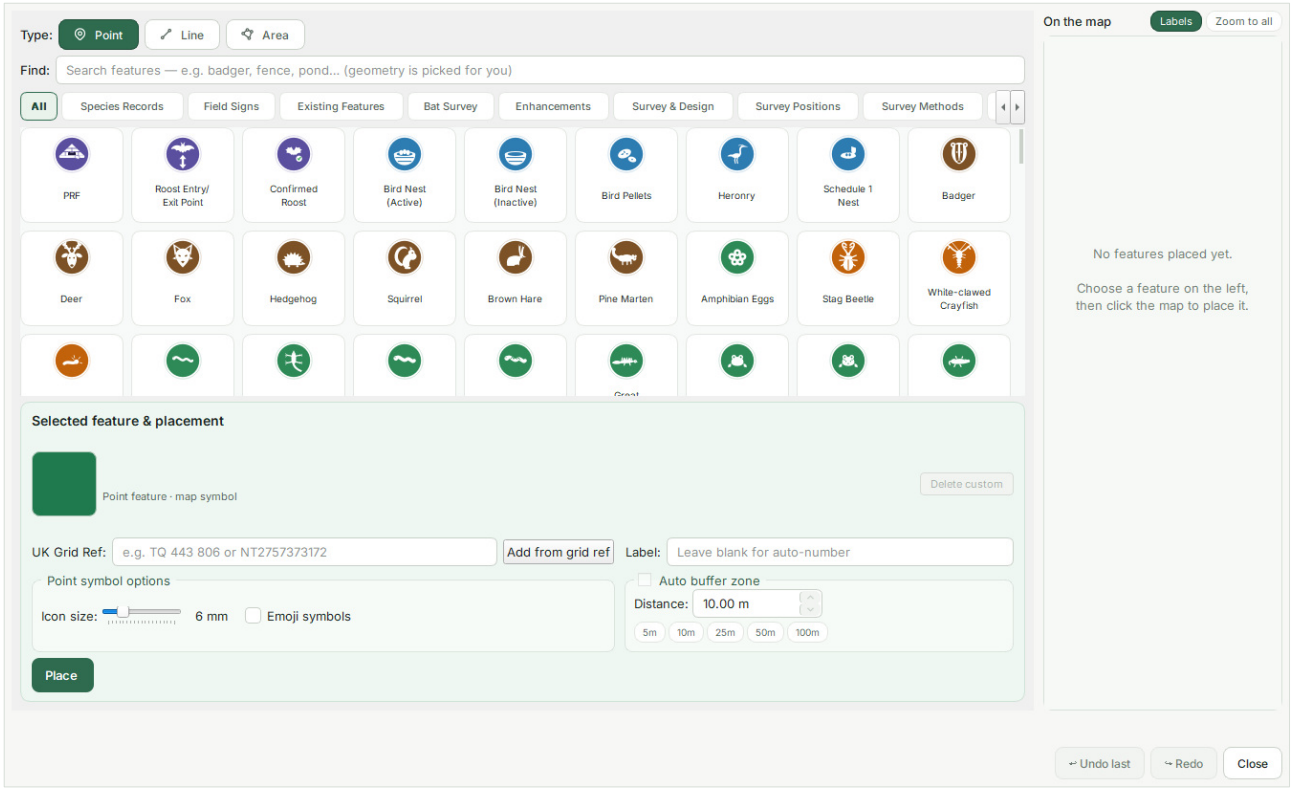
10 Ecological and illustrative features

Everything you record that is not a BNG habitat: species records, field signs, survey infrastructure, enhancements — and presentation-only buildings and trees.

10.1 Feature Mapper

The Feature Mapper records ecological features as points (167 types), lines (20 types) or areas (41 types), each with a dedicated symbol. Features can be auto-numbered (Pond-01, Pond-02 …) or custom-labelled; a master Labels toggle shows or hides every feature label at once; and an optional buffer zone can be drawn around any feature at a set distance.

Survey positions are first-class: observer, camera and thermal camera stations take a bearing and draw a field-of-view cone with a curved label hugging the arc, and the Preliminary Roost Assessment set covers every PRF type with a pink surveyor arrow for vantage directions — feeding the dedicated PRA Map export (chapter 17). Placement is continuous — keep clicking to keep placing — and the full symbol library is reproduced in Appendix B.



The Feature Mapper uses the live ecological symbol catalogue, with point, line and area modes, search, placement options and an on-map feature list.

10.2 Custom feature types

Define your own point, line or area types (name, colour, style, symbol) and they join the pickers, the legends and the Field Mapper catalogue. Custom definitions are stored in your QGIS profile, outside the plugin folder, so plugin updates never touch them.

10.3 Site Plan Visualiser

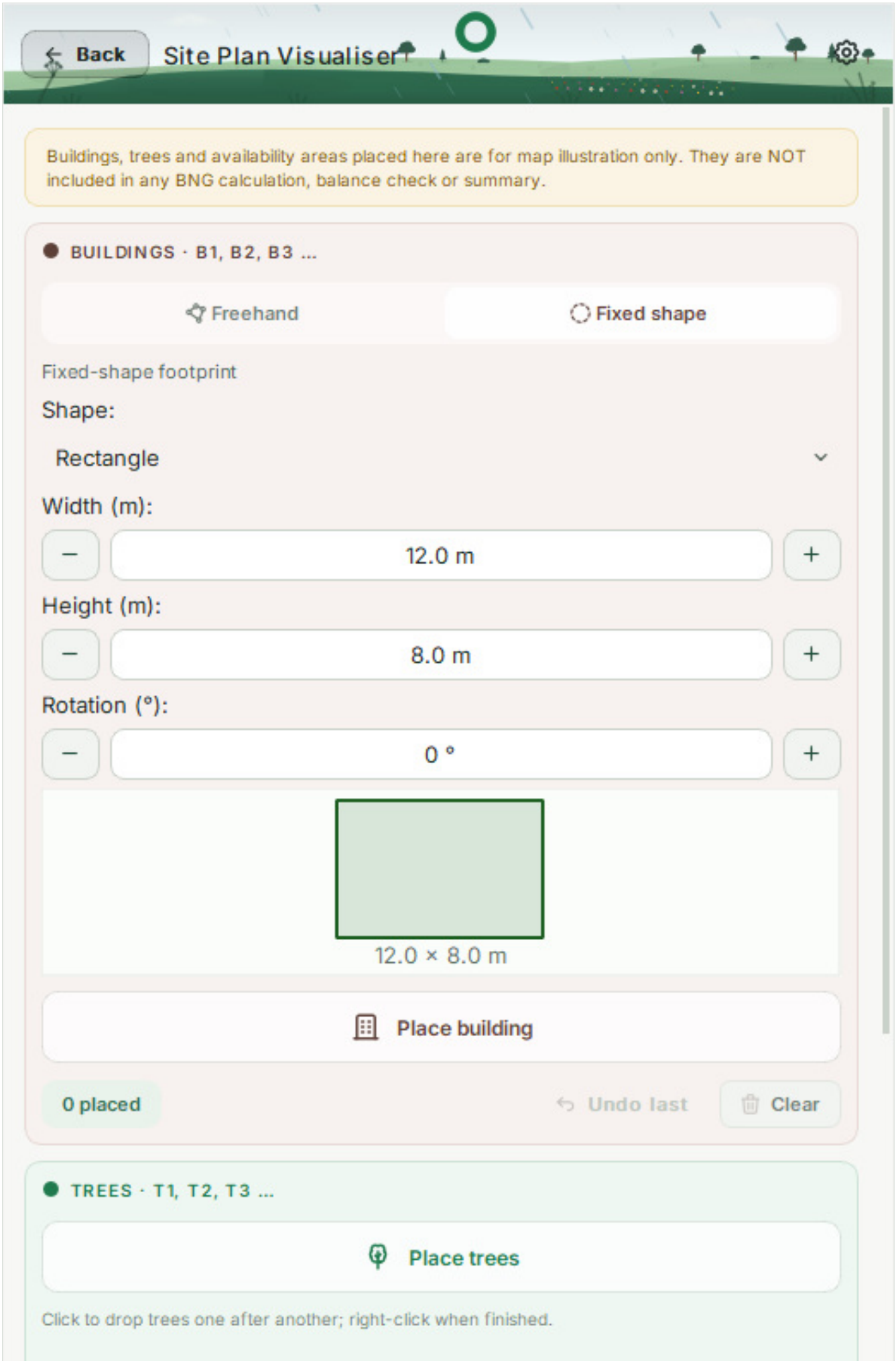
Open Building & Tree Mapper from Habitat Mapping to enter the Site Plan Visualiser. It draws presentation layers for buildings, trees and enhancement availability, then offers dedicated Building, Tree, combined and Enhancement Availability maps. Buildings are auto-numbered B1, B2, B3 … and receive separate colours; trees are labelled T1, T2, T3 … .

> **THE RULE**
>
> These layers are deliberately outside the BNG system: nothing drawn here feeds any calculation or export. Map trees that count towards BNG with the Baseline Trees tool instead (chapter 9).

10.4 Drawing buildings and fixed shapes

- Freehand — click each building corner on the map and right-click to finish. Use this for an irregular footprint or when tracing an architect's georeferenced plan.
- Fixed shape — choose Circle, Ellipse / oval, Square, Rectangle, Stadium / rounded rectangle, Equilateral triangle, Right triangle, Regular polygon or Annulus / ring. Enter the displayed dimensions; regular polygons also take a side count and stadiums a corner radius.
- Placement controls — move the preview with the mouse, scroll to rotate in 15-degree steps, hold Shift while scrolling for 1-degree steps, and use the arrow keys to resize. Left-click places the shape; right-click or Escape finishes.

Use Undo last for the most recent building or Clear to remove the complete presentation layer. The counter at the foot of each card shows how many features have been placed.

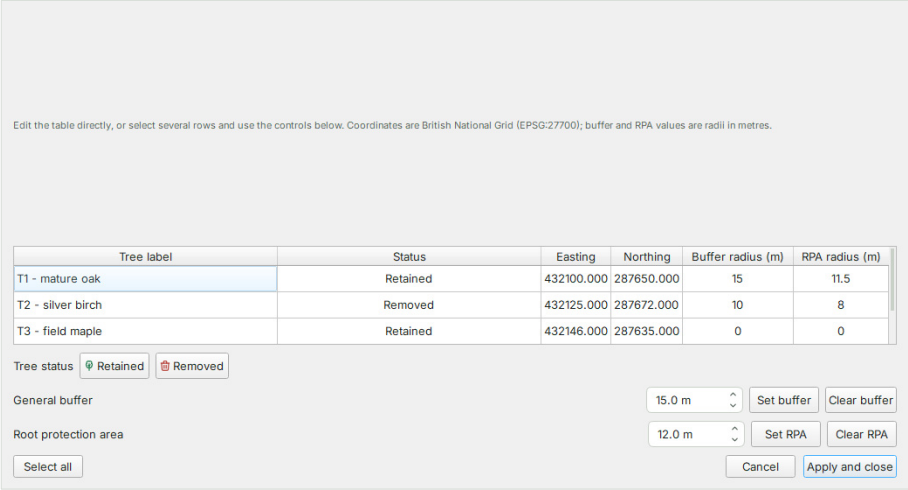

Back
Site Plan Visualiser
Buildings, trees and availability areas placed here are for map illustration only. They are NOT included in any BNG calculation, balance check or summary.
BUILDINGS · B1, B2, B3 ...
Freehand
Fixed shape
Fixed-shape footprint
Shape:
Rectangle
Width (m):
12.0 m
Height (m):
8.0 m
Rotation (°):
0 °
12.0 × 8.0 m
Place building
0 placed
Undo last
Clear
TREES · T1, T2, T3 ...
Place trees
Click to drop trees one after another; right-click when finished.

The Site Plan Visualiser with Fixed shape selected. Dimensions and rotation are entered in metres and degrees before the footprint is placed on the map.

10.5 Tree Mapper and CSV import

Click Place trees, then click repeatedly on the map; right-click or press Escape when finished. For an arboricultural schedule, choose Import trees from CSV. The importer detects British National Grid eastings and northings, latitude and longitude, or OS grid references, shows a placement preview, and imports the accepted rows as labelled illustrative trees.

Choose Manage trees to open the spreadsheet. Each row contains the tree label, Retained or Removed status, easting, northing, general buffer radius and root protection area (RPA) radius. Double-click a cell to edit it, select several rows to apply the same status or radius, or copy a rectangular selection to Excel and paste the edited cells back. Pasting below the last row can add further trees.



The editable tree-management spreadsheet. Coordinates are in British National Grid metres, and buffer and RPA values are radii in metres.

Construction Exclusion Zones and Restricted Activity Zones create cross-hatched overlays from the managed tree information. Each overlay can be shown or hidden and optionally cropped to the red line.

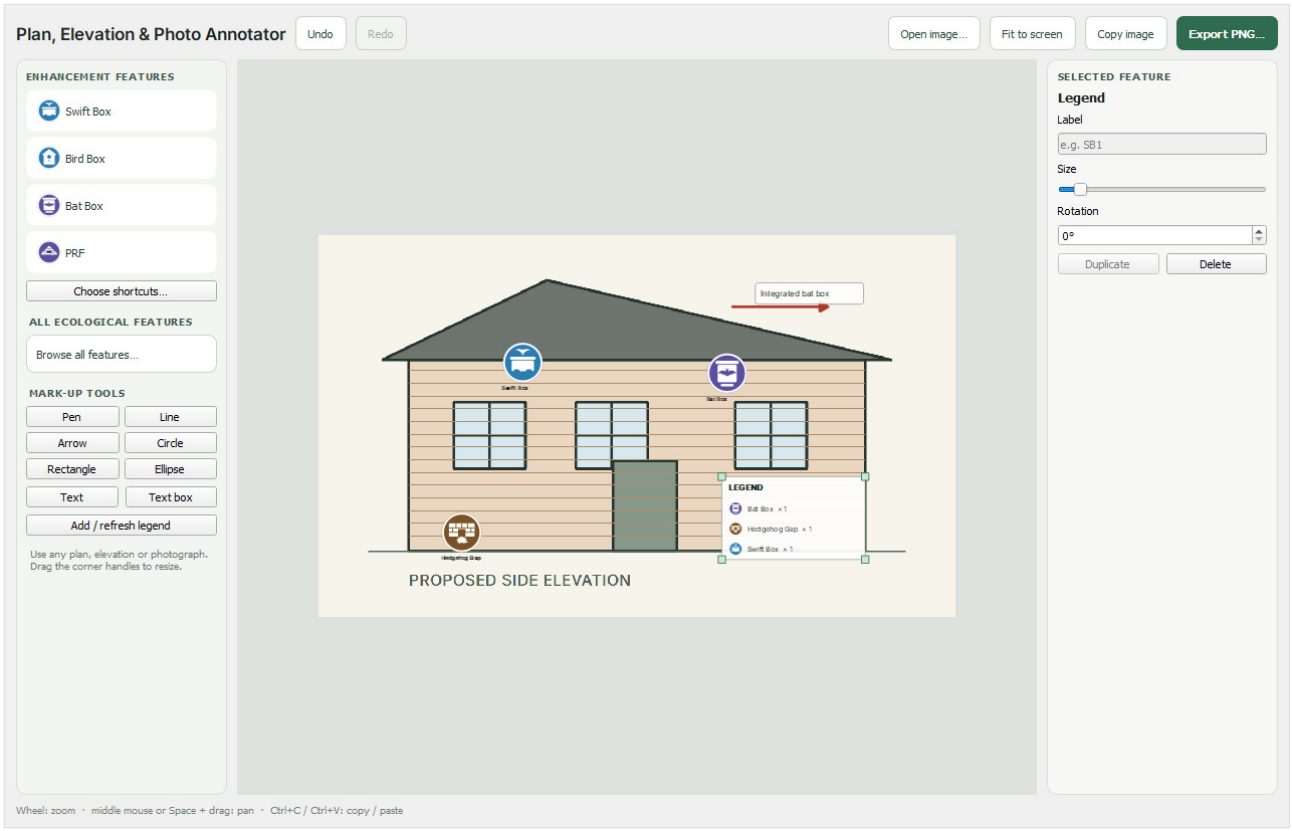
10.6 Enhancement availability

Draw a residential curtilage manually, use Detect from proposal to classify sealed and built proposed habitats as developed, or use Fill remainder as available to mark the remaining red-line area. Where proposed gardens are present, Biome asks whether they form part of the residential curtilage. Re-running detection replaces the old result rather than stacking duplicates.

10.7 Annotating house elevations

The Plan, Elevation & Photo Annotator is primarily for marking up side elevations of houses to show the position and specification of ecological enhancements. Open an elevation drawing, then add enhancement symbols from the full Feature Mapper catalogue, keep frequently used features as shortcuts, and place freehand pen strokes, lines, arrows, circles, rectangles and text. Plans and site photographs can be annotated in the same way.

Select an item to change its label, size, rotation, line colour, thickness, text colour or text-box fill. Items can be dragged, resized, duplicated, deleted, undone and redone. Add / refresh legend builds a key from the enhancement symbols currently on the image. Copy the finished image directly to the clipboard or export it as a PNG for a report or drawing set.



A house side elevation marked up with ecological enhancement symbols, call-outs, a generated legend and editable item properties.

11 The Mobile Field Mapper

A phone web-app served straight from QGIS — record habitats, features, conditions and photos in the field with GPS, then import everything back into the plugin.

11.1 Launching

On the Desk Study tab, Open Field Mapper starts a local server and opens a secure tunnel so GPS works on your phone, then shows a QR code to scan. The first time, the plugin offers to download cloudflared (~30 MB, free and open-source) which creates the tunnel; it downloads once and is reused. Once the page is open on your phone, the map and GPS work in the field.

11.2 What the app carries

The app's catalogue is generated from the live plugin, so it always matches what QGIS can import — every point feature type (167), the full habitat list (157 types) for mapping baseline parcels, hedgerows and watercourses by walking or sketching, the complete condition sheets computed with identical logic, and the GCN Habitat Suitability Index calculator for ponds.

Control	Action
Blue dot	Your live GPS position.
GPS	Pan the map to your current location.
Place Here	Open the feature picker and drop the feature at your GPS point.
Place on Map	Open the picker, then pan a crosshair to place by eye.
Basemap	Double-tap the top bar to cycle Map / Hybrid / Satellite.

11.3 Export and import

Tap the record-count chip, then Export to download a .zip of GeoJSON plus any photos. Back in QGIS, Import Field Survey places every feature — points, lines AND areas — into the correct layers: habitat parcels arrive as baseline habitats with their condition assessments attached, tree records land in the Baseline Trees module, and pond HSI scores import with their ponds. Photos appear on the Photos tab, linked to their features.

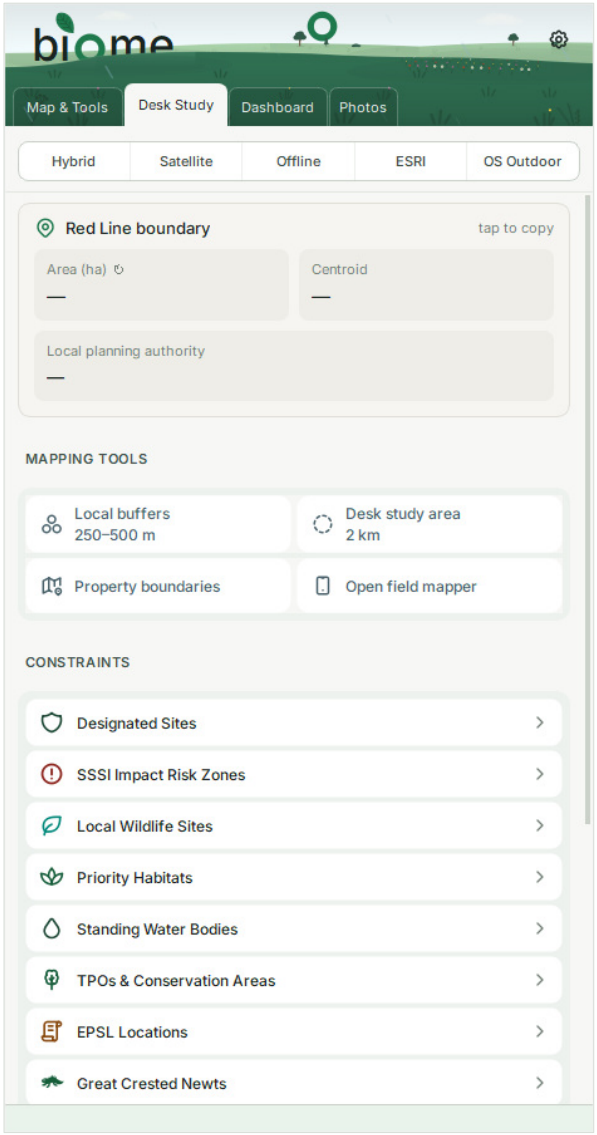
> NOTE
>
> Field-mapped habitats import as BASELINE. Design the proposed scheme in QGIS afterwards — the field survey is your evidence base, not the design surface.

12 The desk study

Live constraint data from national DEFRA and Natural England services — no downloads, no licences — to identify the designations, priority habitats and protected-species risks that shape a scheme.

HOW BIOME HELPS

Missing a constraint — ancient woodland, an SSSI impact-risk zone, a GCN district, a TPO — is a serious and avoidable error, because BNG sits alongside (and never replaces) protected-site and protected-species law. Run the desk study early and design around irreplaceable habitats before they derail a scheme.



The Desk Study tab in QGIS, with basemap choices, boundary summary, mapping tools and expandable constraint searches.

12.1 Mapping tools

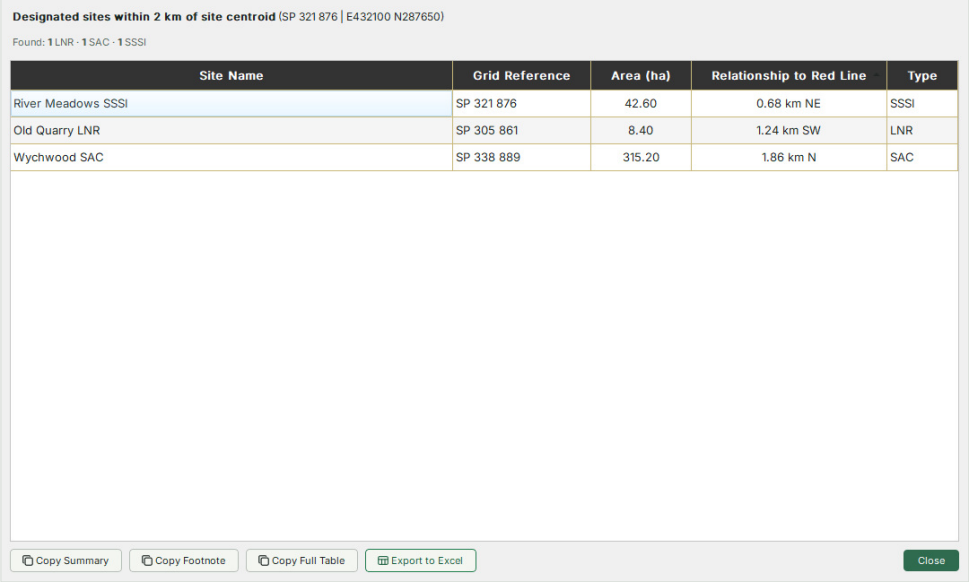
Tool	Use
Redline Buffers	Concentric rings at distances you set (e.g. 50/100/250 m).
Search Buffer	The search radius drawn around the site — every search's radius is configurable in the settings cog.
Feature Info	Click any feature to read its attributes — and add a constraint buffer around just that feature.
Measure from Redline	Click anywhere for the straight-line distance to the nearest red-line edge.
Property Boundaries	HM Land Registry INSPIRE polygons as a live layer.

12.2 Designations, habitats and species

- Designated sites — SSSI, SAC, SPA, Ramsar, NNR, LNR, National Landscapes and National Parks, with SSSI Impact Risk Zones and a distance-tabulated report.
- Priority and irreplaceable habitats — the priority habitat inventory and ancient woodland, the habitats BNG cannot compensate for.
- Protected species licensing (EPSL) — granted licences near the site with species icons, a derived Impact column and dedicated Bats / All-Species report tabs.

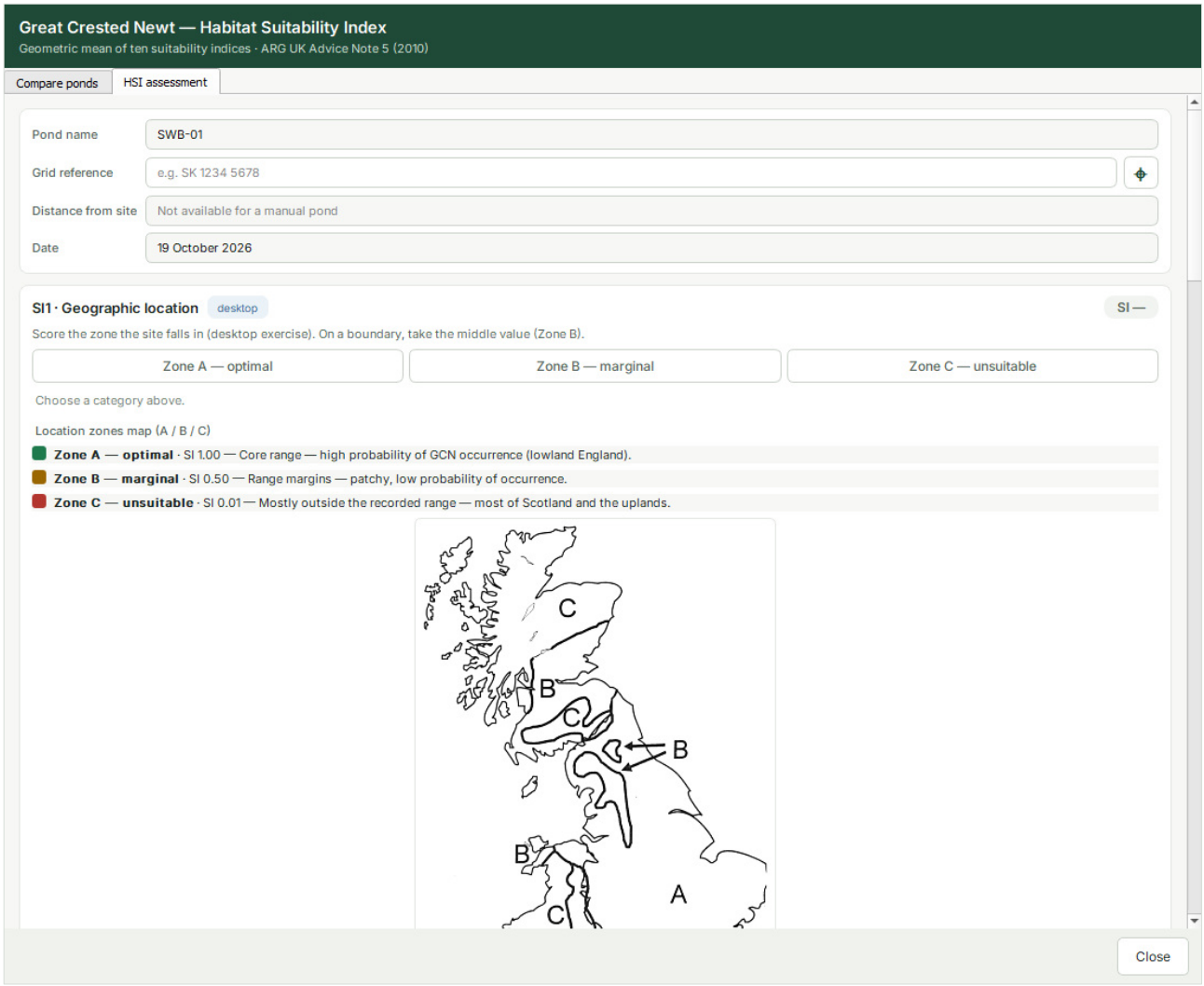
12.3 What the Report buttons do

A Report button runs the relevant spatial query across the current search radius and opens a sortable results window; it does not create a narrative ecology report automatically. Designated Sites, Priority Habitats, EPSL, TPO, Conservation Area and Local Wildlife Site reports use fields suited to their datasets, including distance and direction from the red line where available.



A designated-sites report result. Results can be sorted, copied as a report sentence or formatted table, supplied with a source footnote, or exported to Excel.

- Copy Summary copies a short report-ready statement describing the number of results in the search area.
- Copy Footnote copies the dataset source and expansions of the designation abbreviations actually present.
- Copy Full Table places a formatted table on the clipboard for pasting into the report; its appearance follows the Report table style configured from the settings cog.
- Export to Excel saves the complete result set as an editable spreadsheet. EPSL reports also separate recent and historic records and provide dedicated bats and all-species views.
- Great crested newt — risk-zone mapping for every district Natural England publishes, pond search with numbered water bodies and distance lines, and the GCN Habitat Suitability Index calculator with the standard suitability curves.



The built-in great crested newt HSI assessment, including pond details, factor guidance, suitability controls and reference graphics.

- TPOs — tree preservation orders for most English councils, as points and zones with addresses, species and status, filtered so revoked orders are not reported as live.
- Local knowledge — deep links that open your LPA's own planning map, the Local Nature Recovery Strategy map where one is published, MAGIC, the NBN Atlas and historic mapping, all centred on the site.

12.4 Constraint buffers

Add custom-distance buffer rings around any designated site or priority habitat — whole layers via right-click, or a single feature via its info popup — and the buffer analysis lists what falls inside each ring. Useful for consultation-zone screening and bat-sensitivity distances.

12.5 Survey weather

The Survey Weather pop-out fetches the weather for your survey hours (temperature, wind, rain, cloud) in the format bat-survey reports table it — filling Table 5.0.1 without transcribing from a weather app.

12.6 Desk-study maps

Every constraint set has a one-click print-ready map — designated sites, SSSI IRZ, priority habitats, EPSL, GCN, TPO, Local Wildlife Sites and standing water. Like the habitat maps they are laid out automatically (chapter 17).

12.7 England and Scotland

Biome detects the site region from the red-line centroid and switches the desk-study sources and wording automatically. Use Settings > Site region to force England or Scotland where a site lies close to the border.

Topic	England	Scotland
Designated sites	Natural England and DEFRA services, including National Landscapes	NatureScot and Scottish Government services, including National Scenic Areas
Priority habitats	Natural England Priority Habitats Inventory	NatureScot Habitat Map of Scotland, with EUNIS habitats mapped to Scottish priority-habitat groups
Ancient woodland	Natural England Ancient Woodland Inventory	NatureScot Ancient Woodland Inventory
TPOs	Planning Data and council sources	Spatial Hub Scotland; a free authentication key is required in Biome settings
External map	MAGIC Map	Scotland's environment map
EPSL locations	Natural England licence records	Not available as mapped data; Biome explains the limitation and links to NatureScot's licensing summary

NOTE

The statutory BNG workflow described in this guide applies to England. The Scottish regional switch provides appropriate ecological desk-study data and mapping; it does not imply that England's statutory BNG regime applies in Scotland.

13 The Dashboard

A live control panel for the whole assessment: the headline position, each obligation, trading compliance, credits and data quality — always consistent with the export.

13.1 The hero and obligation list

The hero row shows the site's overall position per module — baseline units, proposed units and percentage change — with an Infographic button that renders the whole dashboard as a shareable PNG poster for clients. Below it, the obligation list breaks the position into the individual things the scheme must deliver, each with a ring gauge showing progress.

13.2 Trading slots

Instead of a single gauge, trading compliance is shown as slots: each loss creates a replacement obligation (“need”) at the right distinctiveness and broad-group tier, and the design's creations and enhancements fill them (“have”). A slot left open tells you exactly what kind of habitat is still missing — which is the question the metric's trading rules actually ask.

13.3 Market and credits

The market table prices the remaining deficit two ways — off-site units at going market rates, and statutory credits at the published tariff — so the cost of not solving a shortfall on-site is visible while the design can still change. Tiering follows the metric's rules, including broad-group replacement for Medium losses and the offset cascade between tiers.

13.4 Data quality

The Dashboard flags features missing a condition or sub-type, unmapped ground inside the red line, and other issues that would weaken the export — each warning names the features so you can fix them from Manage.

> **THE RULE**
>
> On-screen units are indicative — computed from QGIS geometry to guide design and catch errors. The planning submission always uses the exported statutory workbook (chapter 18).

14 The Enhance tab

Turns a shortfall into actions: what to enhance on-site, and what the off-site routes would take.

14.1 On-site

The on-site view reads your retained baseline and recommends enhancement moves — which parcels can move up in condition or to an eligible better habitat, what each move is worth, and whether it is feasible for that habitat (the engine checks each move against the metric's enhancement rules and the 30-year timeframe, capping what a habitat can realistically reach). Moves that would score NEGATIVELY after the temporal discount — slow high-distinctiveness habitats — are flagged “retain instead” rather than recommended.

14.2 Off-site routes

The off-site view compares three routes for the remaining gap, all scored under the off-site rules (spatial risk multiplier, the same trading tiers, and the correct creation-vs-enhancement temporal discounts):

> NOTE
>
> Since 6 August 2026, a qualifying minor development that is not exempt may consider on-site and registered off-site delivery at the same time and may choose off-site delivery without first exhausting on-site options. Major developments retain the normal on-site then off-site sequence. Statutory credits remain the last resort in both cases.

- Your own land — score blue-line land outside the red line exactly as the metric's off-site sheets would.
- A habitat bank — the units to buy, tiered so the purchase actually satisfies the trading rules, not just the number.
- Statutory credits — the last-resort tariff, priced from the current published schedule.

14.3 Watercourses too

Watercourse shortfalls route through the same views with the watercourse module's own rules — including encroachment-based enhancement — so a rivered site gets real recommendations, not an area-shaped approximation.

15 Managing what you've mapped

Manage Features is the single place to review, restyle, edit, duplicate, trim and delete everything you have placed.

15.1 The feature list

Features are grouped by layer with per-group fold-out sections; hovering a row highlights the feature on the map. Each hedgerow and tree row carries a live status badge — Retained, Enhanced, Created or Lost — computed by the same classifier as the export, so the badge is always the truth.

15.2 Row actions

Button	Does
Eye	Show / hide the feature on the map.
Palette	Per-feature colour override — keeps the symbology but recolours one feature (useful where neighbours share a colour). Reset restores the bundled style.
Clipboard	Open, copy or export the condition sheet (assessable habitats).
Pencil	Edit type, condition and other properties.
Copy	Duplicate into the paired baseline/proposed layer as a retained twin (chapter 8).
Scissors	Trim a section out of a hedgerow or watercourse (chapter 8).
Trash	Delete — recoverable this session from the deleted-features bin.

15.3 Moving and reshaping

Click-to-move relocates point features; reshape re-draws a geometry with a green Finish / red Cancel in the message bar. Every destructive action lands on the global undo stack (chapter 3.2), so nothing here is a one-way door.

16 Measurements and de minimis

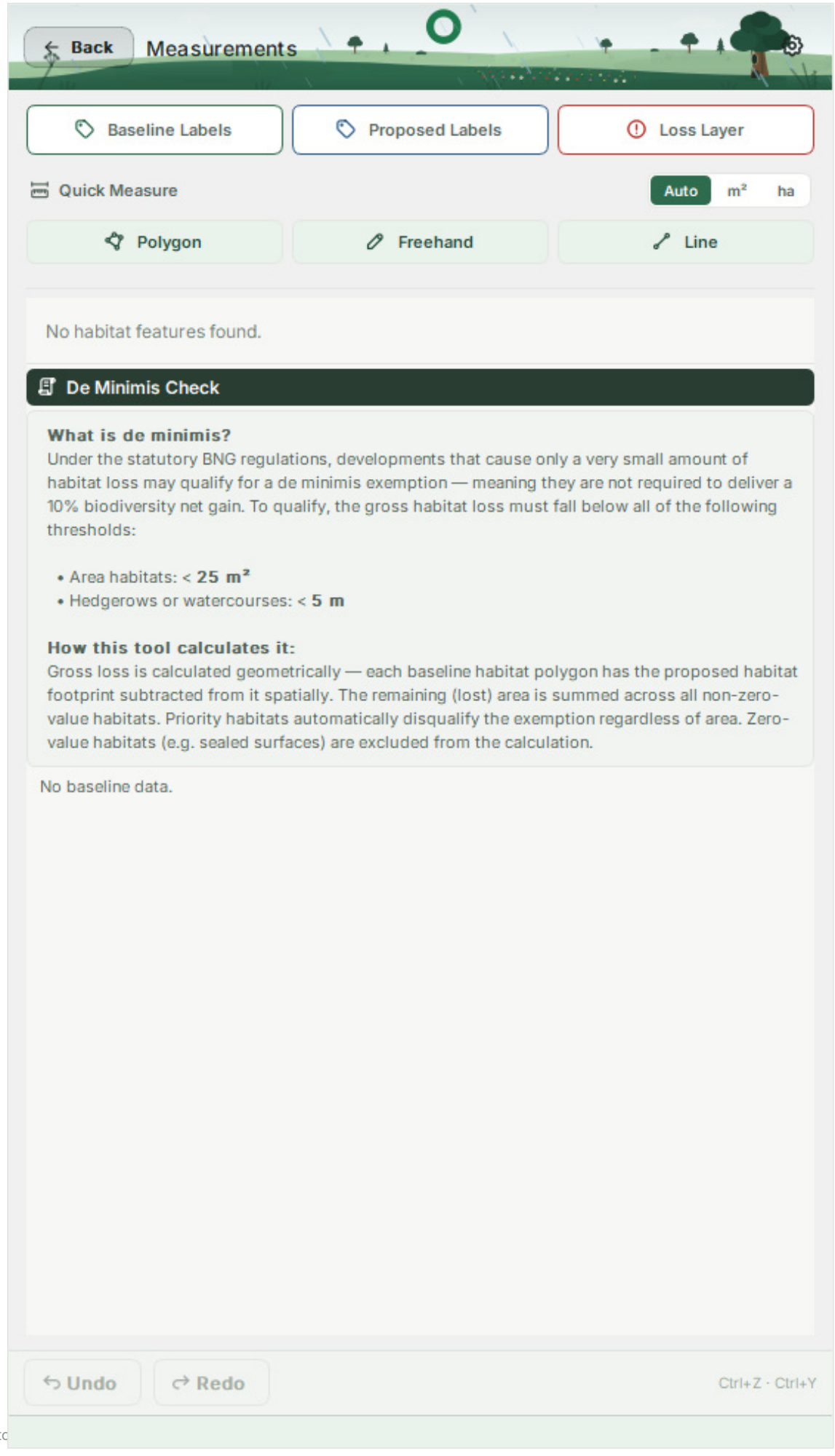
Every figure Biome shows is a true ground measurement — and the de minimis exemption is screened with evidence, not by eye.

16.1 Ground-true measurement

Web-mapping projections inflate naive areas by roughly 2.6x at UK latitudes (and lengths by 1.6x). Biome never uses planar maths for anything you see: areas and lengths are measured ellipsoidally on the ground, printed scales are true scales, and the scale bar segments are sized accordingly. If another tool disagrees with Biome's areas, check whether it is measuring in projected units.

16.2 The Measurements dialog

Measurements toggles per-feature area/length labels for any layer, offers a quick measure-sketch tool with a live cursor counter and an m2/ha unit toggle, and overlays a loss layer highlighting baseline areas not covered by a matching proposed habitat.


Back
Measurements
Baseline Labels
Proposed Labels
Loss Layer
Quick Measure
Auto
m²
ha
Polygon
Freehand
Line
No habitat features found.
De Minimis Check
What is de minimis?
Under the statutory BNG regulations, developments that cause only a very small amount of habitat loss may qualify for a de minimis exemption — meaning they are not required to deliver a 10% biodiversity net gain. To qualify, the gross habitat loss must fall below all of the following thresholds:
• Area habitats: < 25 m²
• Hedgerows or watercourses: < 5 m
How this tool calculates it:
Gross loss is calculated geometrically — each baseline habitat polygon has the proposed habitat footprint subtracted from it spatially. The remaining (lost) area is summed across all non-zero-value habitats. Priority habitats automatically disqualify the exemption regardless of area. Zero-value habitats (e.g. sealed surfaces) are excluded from the calculation.
No baseline data.
Undo
Redo
Ctrl+Z · Ctrl+Y

The Measurements screen provides baseline and proposed labels, the loss overlay, quick polygon, freehand and line measuring, unit selection, and the de minimis check in one place.

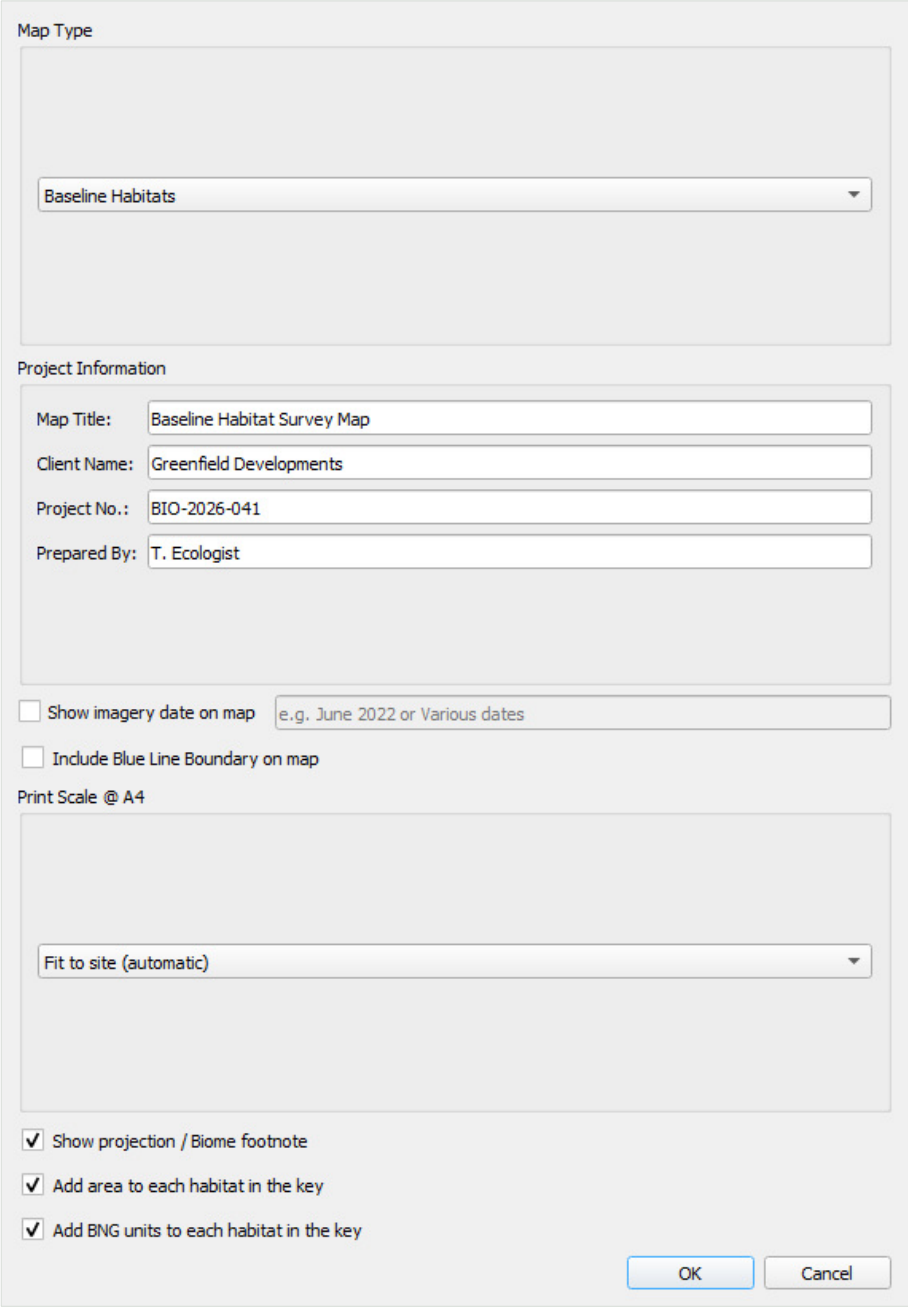
16.3 De minimis screening

> COMMON BNG PITFALL
>
> The de minimis exemption requires no impact on on-site priority habitat and impacts of less than 25 m2 of on-site area habitat and less than 5 m of on-site linear habitat. It is distinct from the 0.2 ha site-area exemption introduced on 6 August 2026.

Biome computes the geometric gross loss against the statutory thresholds and produces a De Minimis Assessment Map, so an exemption claim is documented, or ruled out, with numbers. Check priority habitat separately before relying on the result. For the 0.2 ha exemption, use the complete planning red-line area rather than the mapped loss area.

17 Maps, layouts and 3D exports

One-click print-ready maps that lay themselves out — plus clipboard export, georeferenced architects' plans and two 3D site viewers.



The Final Map popup for a baseline habitat map. Options change with the selected map type, so only relevant controls are shown.

17.1 Final Map

Click Final Map, choose a map type, complete the project fields and press OK. Biome creates and opens a QGIS print layout rather than immediately saving a flat image. The result is an A4 landscape sheet with the map filling the page and a themed sidebar carrying the title, legend, project metadata, north arrow and scale bar. Inspect or adjust the layout before using QGIS's Export as Image or Export as PDF action.

17.2 Every control in the Final Map popup

Control	What it does
Map Type	Selects the specialist layer set, title and legend rules described below.
Map Title	Starts with the default for the selected type but can be edited. The change is remembered for that type until QGIS closes.
Client Name / Project No.	Printed in the sidebar and remembered with the QGIS project. Shared with the Filled Metric's Applicant and Project name fields.
Prepared By	Printed in the sidebar and remembered for later maps.
Show imagery date	Adds the typed aerial-imagery date or wording such as 'Various dates'. Leave unticked where it is unknown.
Include Red Line Boundary	Available on specialist ecological, emergence, building and tree maps. Baseline, Proposed and De Minimis maps always include it.
Include Blue Line Boundary	Adds the wider ownership or off-site boundary. Custom maps control this through their checklist.
Print Scale @ A4	Fit to site frames the red line. Fixed true-ground scales are 1:100, 1:200, 1:500, 1:1,250, 1:2,500, 1:5,000 and 1:10,000; an unsuitable fixed scale can crop the site.
Projection / Biome footnote	Shows or hides the small CRS and generator line at the foot of the sidebar.
Add area to each habitat	Baseline and Proposed only; adds polygon area or linear length to each legend entry.
Add BNG units	Baseline and Proposed only; adds the same indicative unit figures shown on the Dashboard.
Survey Date(s)	Emergence Survey Layout only; up to three ticked dates print on separate lines.
Include georeferenced layer	Appears when aligned architects' plans or other rasters are present. Tick each overlay required above the basemap.
Layers to Include	Custom only. Tick individual layers, or a group to toggle all its children. Basemaps follow the visible map state.

17.3 What each Map Type submenu produces

Map type	Shows
Baseline Habitats	Baseline polygons, hedgerows, watercourses and individual trees.
Proposed Habitats	Proposed polygons, hedgerows, watercourses and individual trees.
De Minimis Assessment	Baseline, proposed and net loss for exemption screening.
PRF and Target Note Map	PRF points, water bodies, bird nests and target notes.
PRA Map	Every Preliminary Roost Feature type, surveyor vantage arrows and target notes.
Ecological Enhancement Map	Boxes, bricks, banks, gaps and planting zones, legended by type.
Emergence Survey Layout	Surveyors, stations, cameras, detectors, field-of-view wedges, buildings, trees and survey dates.
Building Map	Illustrative buildings, with each numbered building colour shown in the key.
Tree Map	Illustrative trees labelled T1, T2 … with tree-management overlays.
Building & Tree Map	The two presentation layers together.
Enhancement Availability Map	Curtilage and developed land compared with land available for enhancement.
Custom	Any combination of project layers selected from a grouped checklist.
Desk-study maps	The equivalent one-click map within each constraint section in chapter 12.

17.4 What happens after pressing OK

Biome chooses the layers and render order, frames the site, creates the sidebar and builds the legend from features that are actually present. It registers the result in the QGIS project and opens the Layout Designer. Re-running a map creates a new layout; it does not silently overwrite an earlier layout.

- Extent frames itself to the red line; titles are editable and have persistent defaults available through the settings cog.
- The legend builds itself from the included layers and fits with a ladder of one column, two columns and, where necessary, reduced per-feature figures.
- Scales are true: the printed 1:X and scale bar use ground scale, not an inflated Web-Mercator denominator. Ctrl+scroll fine zoom snaps to round printed scales.

17.5 Getting maps out

Every layout designer gains a Copy to Clipboard action, and the native Export as Image copies the saved file automatically, so a generated map can be pasted straight into a report. The usual QGIS PDF export remains available for issue-quality mapping.

17.6 Architects' plans, imagery and 3D

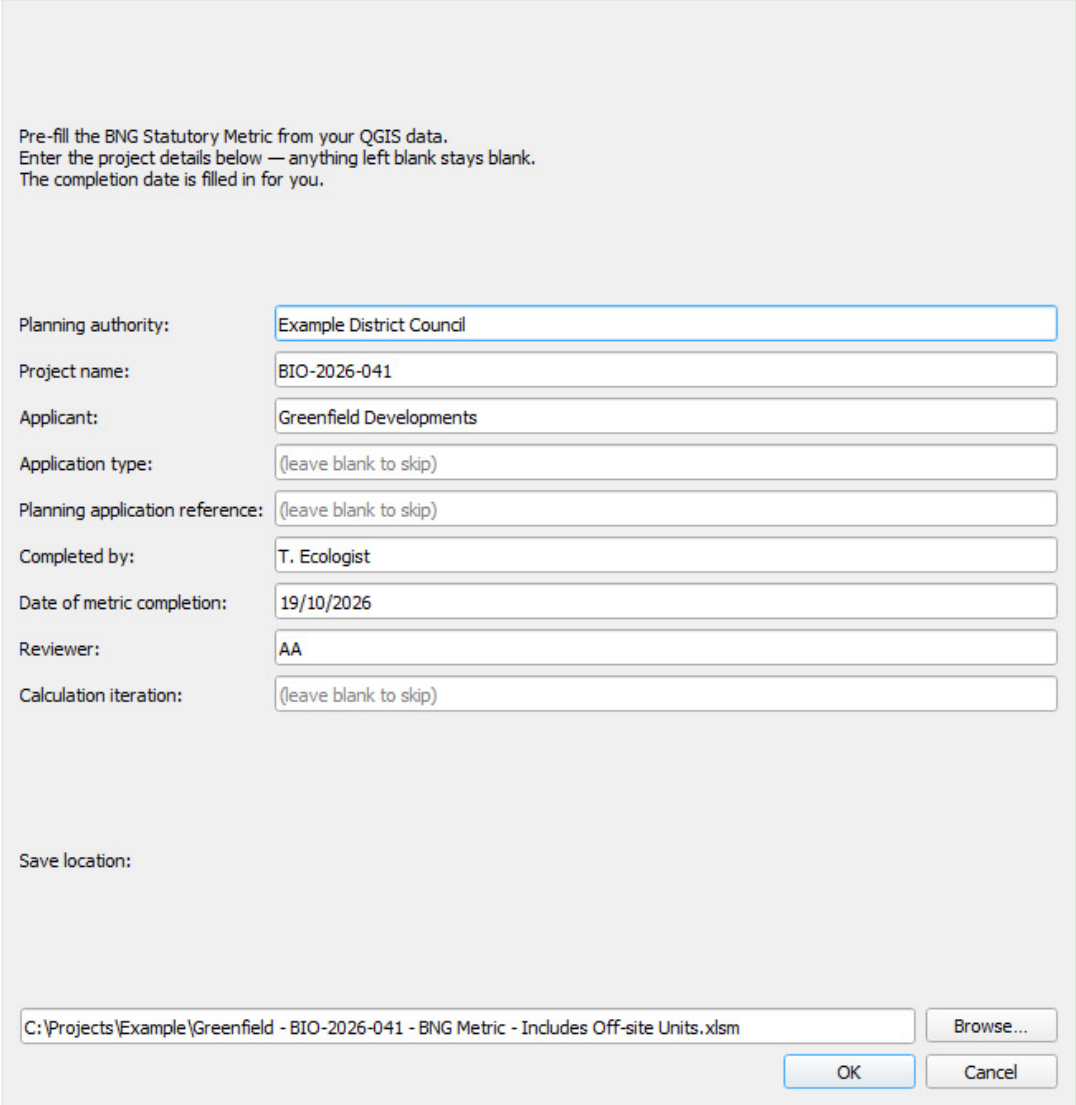
- Georeferenced plans and overlays — architects' plans aligned with the Georeferencer can be included on any Final Map. The same workflow supports drone orthophotos and suitable imagery.
- Google Earth spotlight — exports a KML that dims everything outside the red line in Google Earth's 3D view.
- 3D Site Spotlight — a self-contained Cesium viewer with photorealistic 3D tiles, classification fog outside the site and a boundary ribbon.

18 Exporting the metric and tables

The endpoint of the workflow: the statutory workbook, pre-filled and consistent with everything the panel showed you.

18.1 Blank vs Filled Metric

- Blank Metric — a clean, unmodified copy of the statutory workbook to fill in by hand.
- Filled Metric — the statutory workbook pre-filled from your mapping: area, hedgerow and watercourse baselines, creations and enhancements, with enhancement rows linked to their baselines and off-site entries routed to the off-site sheets.



The Filled Metric export collects the Statutory Metric project details and save location. Blank fields remain blank in the workbook.

18.2 How the pre-filled metric works

Biome starts with its bundled macro-enabled Statutory Biodiversity Metric workbook and writes only to user-input cells. It preserves the statutory formula cells, dropdown logic, macros, charts and headline calculations. Project details go to the Start sheet; the completion date is filled with today's date but remains editable.

- Baseline sheets receive mapped baseline area habitats, hedgerows and watercourses, including measured area or length, condition, strategic significance and retained or enhanced quantity.
- Creation sheets receive proposed features that are not retained, together with the proposed habitat, condition, strategic significance and available delivery fields.
- Enhancement sheets link back to their source baseline rows. The workbook spawns them from enhanced baseline quantities, and Biome supplies the proposed state needed to score the uplift.
- Individual trees are written into the area module using the statutory area equivalent for their mapped size class; there is no separate tree module.
- Off-site delivery within the blue line can be included or excluded when prompted. Included habitat is routed to the D, E and F sheets with its reference and spatial-risk category, rather than being mixed into on-site A, B and C.

Before writing, Biome checks geometry and validates habitat names and required fields against the workbook. Resolve warnings rather than assuming an unrecognised or blank entry will score. The suggested filename includes the applicant, project name, date and an off-site marker where applicable; the last folder is remembered per project.

> **THE RULE**
>
> The exported .xlsm is the source of truth for a submission. Open it in Excel, let it recalculate and review the Headline Results before submitting. The workbook always supersedes an on-screen figure.

18.3 Reviewing the exported workbook

Open the .xlsm in desktop Excel, enable its content if prompted and allow recalculation. Check the Start sheet, every populated baseline, creation and enhancement sheet, the trading summaries and Headline Results. Complete statutory fields that cannot be derived from the mapping. If the mapping changes, export again rather than manually trying to synchronise an older workbook.

18.4 Export Table

Export Table writes every mapped habitat to an Excel reference workbook laid out by module, including type, condition, ground-true area or length, distinctiveness, strategic significance and indicative units. Completed condition assessments can also be exported in the statutory condition-sheet format, one sheet per assessed parcel, ready for the report appendix.

19 Data, sharing and troubleshooting

Where everything lives, how to hand a project over, and the quick answers.

19.1 Where your data lives

All habitat and ecological data sits in the project's GeoPackage (<project>_bng_data.gpkg); layer styles, groups and settings live in the QGIS project file. Session-only helpers — basemaps, desk-study service layers, distance lines — are deliberately stripped from saves, so a project reopened on another machine never chases missing services; persistent survey layers (water bodies included) are saved in the GeoPackage. To share a project, zip the QGIS project file and the .gpkg together.

> NOTE
>
> Edit the GeoPackage only through Biome's tools. If you must edit externally, the settings cog's live-refresh option reloads Biome layers automatically when the file changes.

19.2 Pre-submission checklist

Check	Where in Biome
No unclassified gaps inside the red line	Dashboard coverage check
No overlapping habitat polygons	Tools > Overlaps
Each module hits +10% on its own	Dashboard hero
Every trading slot filled	Dashboard trading slots
Losses of changed parcels accounted for	Net-difference classifier (automatic)
Hedgerow/watercourse fates honest	Manage badges; pair links; trimmer
Condition evidenced, not optimistic	Condition sheets; Photos tab
Constraints & irreplaceable habitats checked	Desk Study tab
De minimis claims documented	De Minimis Assessment Map
0.2 ha exemption checked against whole red-line area	Red-line area summary; application date and priority habitat checked separately
Workbook recalculated in Excel	Filled Metric export

19.3 Troubleshooting

Symptom	Resolution
Features not visible after import	Right-click the layer > Zoom to Layer.
GPS not working on the phone app	Ensure cloudflared downloaded; the landing page should confirm the secure tunnel.
Photos missing from the Photos tab	Use the .zip field export — bare .geojson does not carry photos.
Units differ from the metric workbook	Expected — on-screen units are indicative; rely on the Filled Metric.
Areas differ from another GIS	Biome measures on the ground (ellipsoidal); check the other tool isn't using projected planar units (chapter 16).
A button seems armed after switching tabs	Biome disarms stranded map tools automatically; press Escape or the Pan tool if an older QGIS misbehaves.

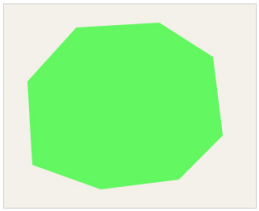
NOTE

Biome is independent software that supports, but does not replace, professional ecological judgement or the statutory metric. Confirm BNG requirements, exemptions and metric versions against current GOV.UK guidance for every application.

A Habitat symbology reference

Every habitat type, rendered with the exact map symbology the plugin uses — 131 area habitats, 13 hedgerows, 5 watercourses and the tree size classes.

Grassland


Modified grassland

Lowland meadows

Lowland calcareous grassland

Lowland dry acid grassland

Other neutral grassland

Other lowland acid grassland

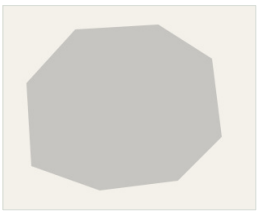
Upland hay meadows

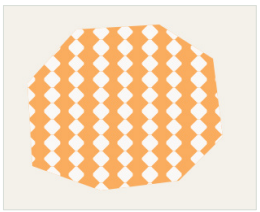
Upland calcareous grassland

Upland acid grassland

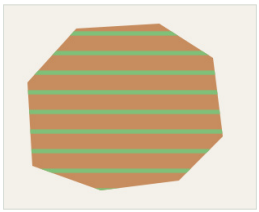
Floodplain wetland mosaic and CFGM

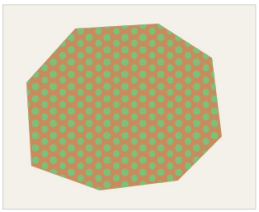
Bracken

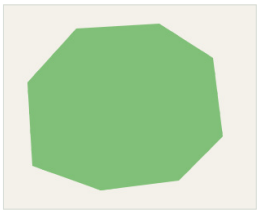

Tall herb communities (H6430)

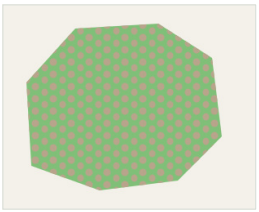

Traditional orchards

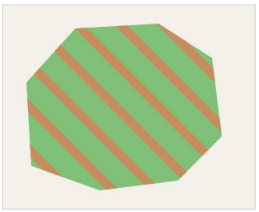
Woodland


Lowland mixed deciduous woodland


Other woodland; mixed

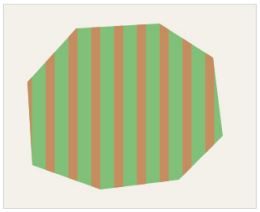

Lowland beech and yew woodland

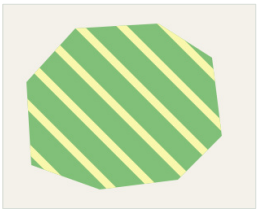

Other woodland; broadleaved

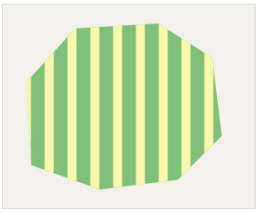

Wet woodland


Upland oakwood

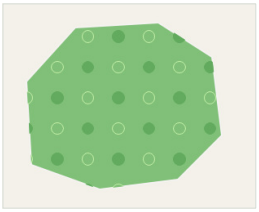

Upland mixed ashwoods


Upland birchwoods


Native pine woodlands


Other Scot's pine woodland


Other coniferous woodland

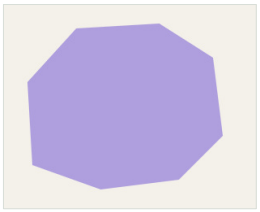

Wood-pasture and parkland

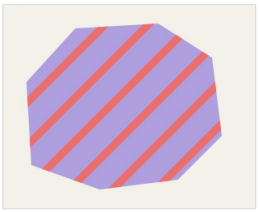
Felled

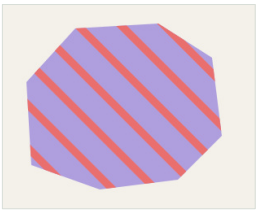
Heathland & Shrub

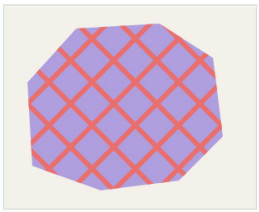

Lowland heathland

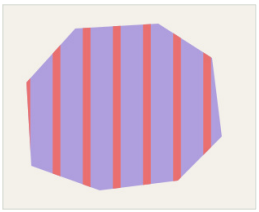

Upland heathland

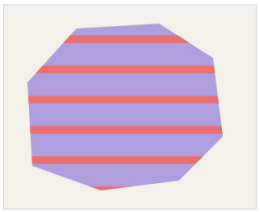

Mountain heaths and willow scrub

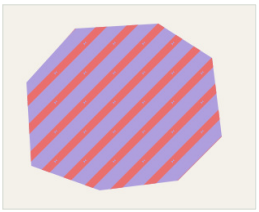

Mixed scrub

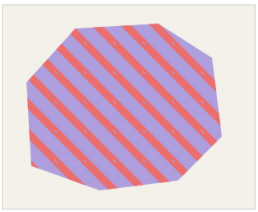

Blackthorn scrub

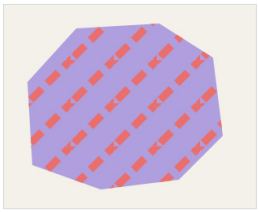

Bramble scrub

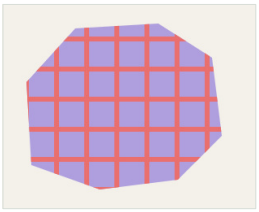

Hawthorn scrub

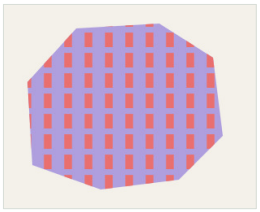

Hazel scrub


Gorse scrub

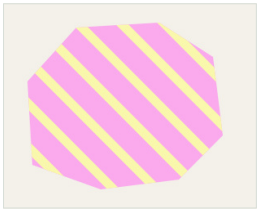

Willow scrub


Rhododendron scrub


Dunes with sea buckthorn (H2160)

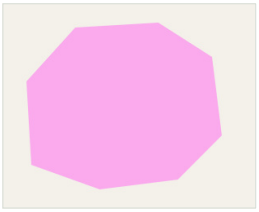

Other sea buckthorn scrub

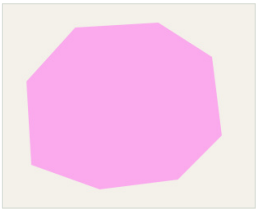
Wetland

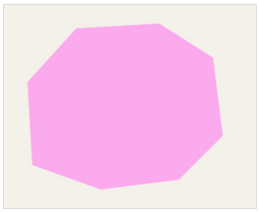

Fens (upland and lowland)


Blanket bog


Lowland raised bog


Transition mires and quaking bogs (H7140)


Depressions on peat substrates (H7150)


Oceanic valley mire[1] (D2.1)

Purple moor grass and rush pastures


Reedbeds

Lakes

Ponds (priority habitat)

Ponds (non-priority habitat)

Marl lakes

High alkalinity lakes

Moderate alkalinity lakes

Low alkalinity lakes

Peat lakes

Aquifer fed naturally fluctuating water bodies

Reservoirs

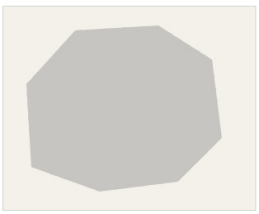
Temporary lakes ponds and pools (H3170)

Ornamental lake or pond

Watercourse footprint

Sparsely Vegetated Land

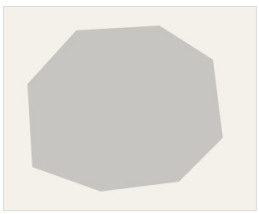

Inland rock outcrop and scree habitats


Other inland rock and scree

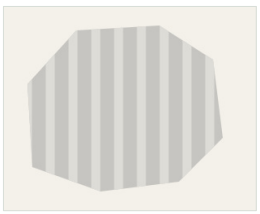

Limestone pavement

Calaminarian grasslands


Ruderal/Ephemeral


Tall forbs


Coastal sand dunes


Coastal vegetated shingle

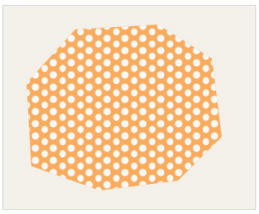

Maritime cliff and slopes

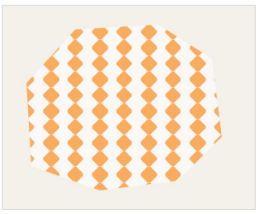
Cropland


Cereal crops

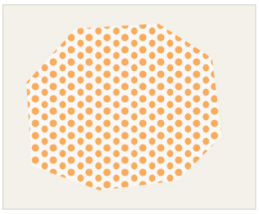

Cereal crops other

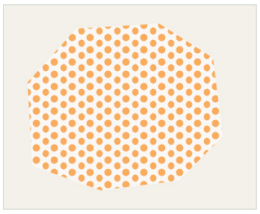

Winter stubble

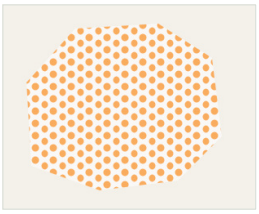

Non-cereal crops

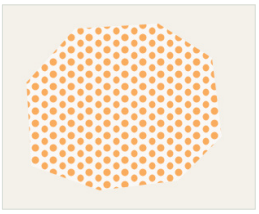

Horticulture

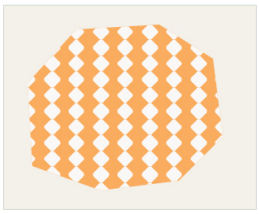

Temporary grass and clover leys


Arable field margins cultivated annually


Arable field margins game bird mix

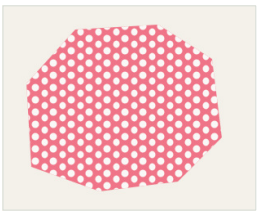

Arable field margins pollen and nectar

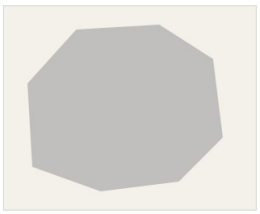

Arable field margins tussocky

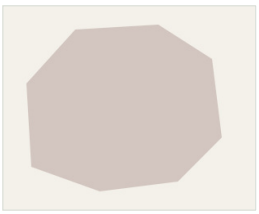

Intensive orchards

Urban


Developed land; sealed surface

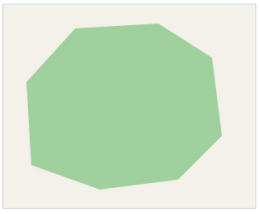

Artificial unvegetated, unsealed surface

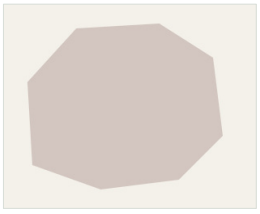

Built linear features

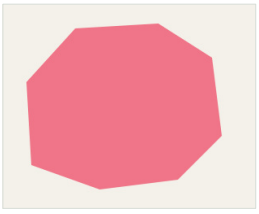

Vacant or derelict land

Open mosaic habitats on previously developed land

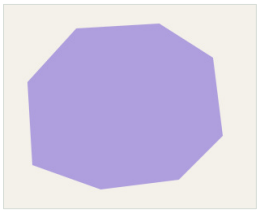
Cemeteries and churchyards

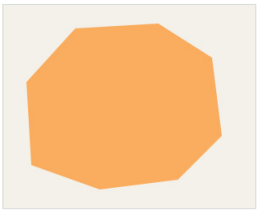

Vegetated garden


Unvegetated garden


Actively worked sand pit quarry or open cast mine

Bare ground


Introduced shrub


Allotments

Biodiverse green roof

Intensive green roof

Other green roof

Facade-bound green wall

Ground based green wall

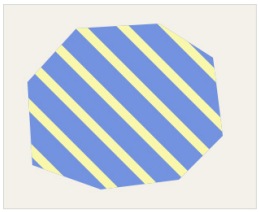
Ground level planters

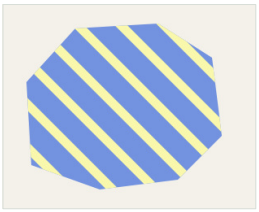
Bioswale

Rain garden

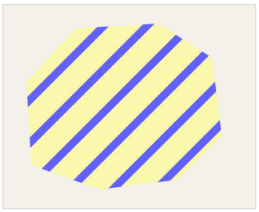
Sustainable drainage system

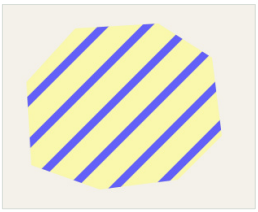
Coastal & Intertidal

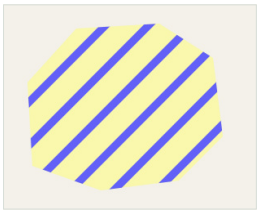

Saltmarshes and saline reedbeds

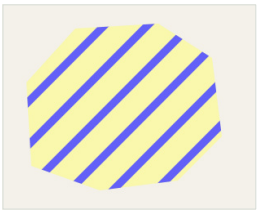

Artificial saltmarshes and saline reedbeds

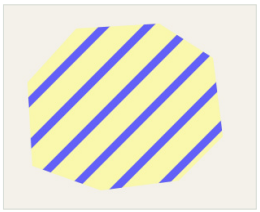
Coastal lagoons

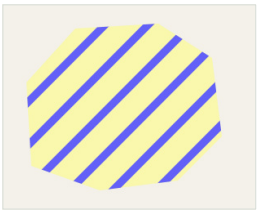

Littoral mud

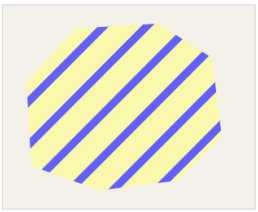

Littoral sand

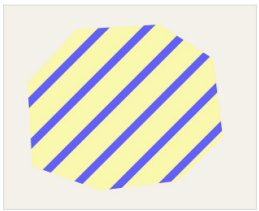

Littoral coarse sediment

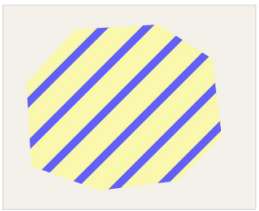

Littoral mixed sediments

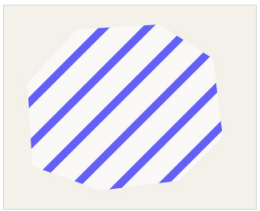

Littoral muddy sand

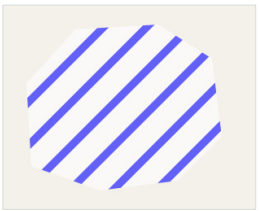

Littoral seagrass

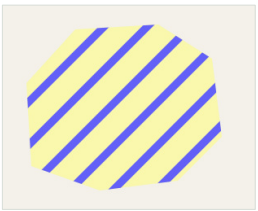

Littoral seagrass on peat, clay or chalk

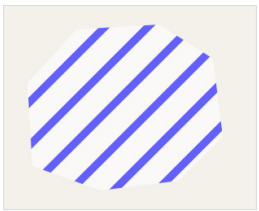

Littoral biogenic reefs - Mussels

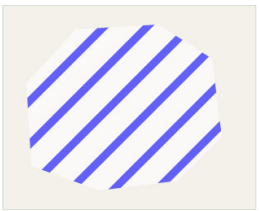

Littoral biogenic reefs - Sabellaria

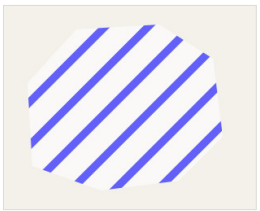

Features of littoral rock

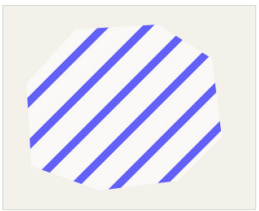

Features of littoral rock - on peat, clay or chalk

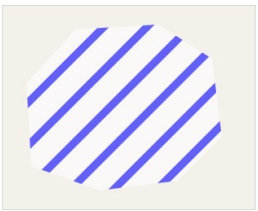

Features of littoral sediment

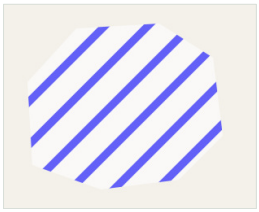

High energy littoral rock

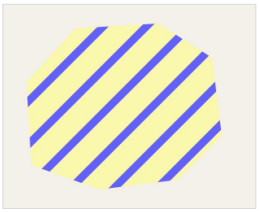

High energy littoral rock - on peat, clay or chalk

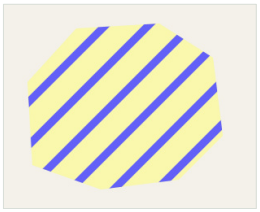

Moderate energy littoral rock

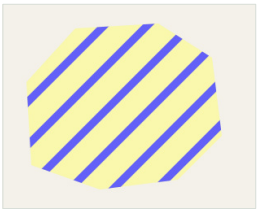

Moderate energy littoral rock - on peat, clay or chalk

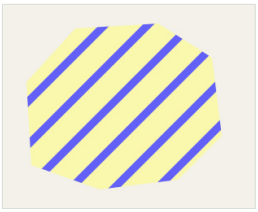

Low energy littoral rock

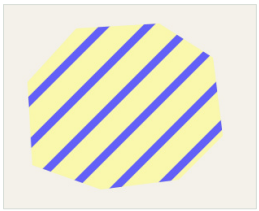

Low energy littoral rock - on peat, clay or chalk

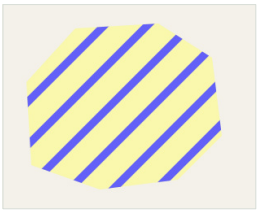

Artificial littoral biogenic reefs

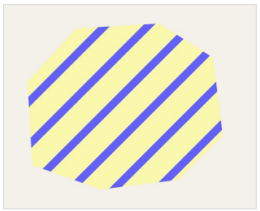

Artificial littoral coarse sediment

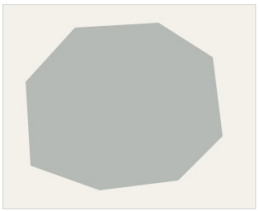

Artificial littoral mixed sediments

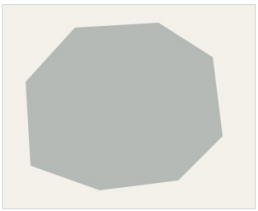

Artificial littoral mud

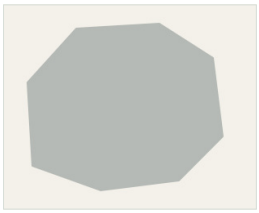

Artificial littoral muddy sand


Artificial littoral sand

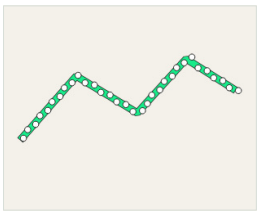

Artificial littoral seagrass


Artificial hard structures

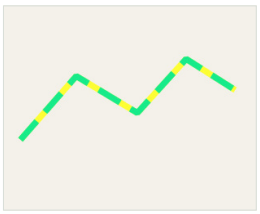

Artificial features of hard structures

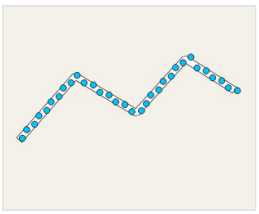

Artificial hard structures with integrated greening of grey infrastructure (IGGI)

Watercourse (line symbology)


Other rivers and streams

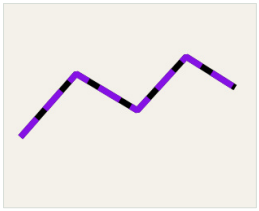

Priority habitat

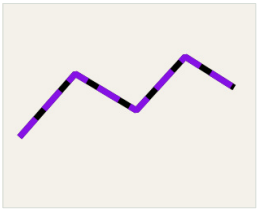

Ditches

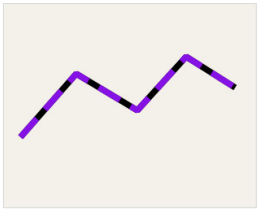

Canals

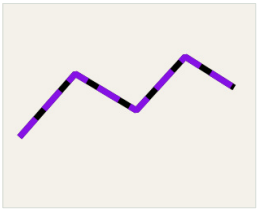

Culvert

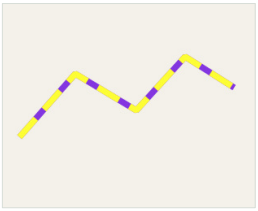
Hedgerow (line symbology)

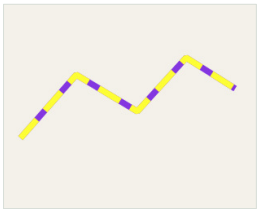

Species-rich native hedgerow

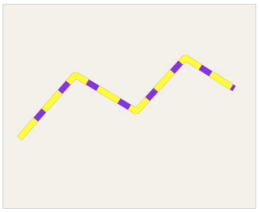

Species-rich native hedgerow with trees

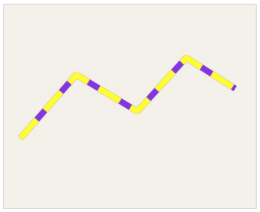

Species-rich native hedgerow - associated with bank or ditch

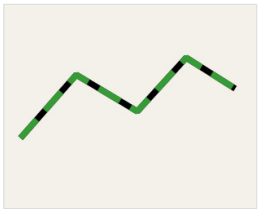

Species-rich native hedgerow with trees - associated with bank or ditch

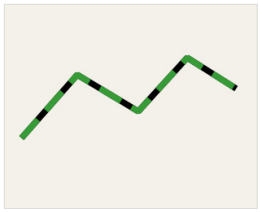

Native hedgerow

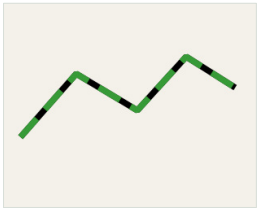

Native hedgerow with trees

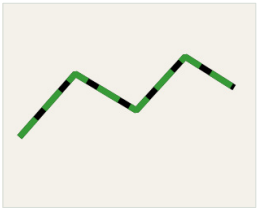

Native hedgerow - associated with bank or ditch

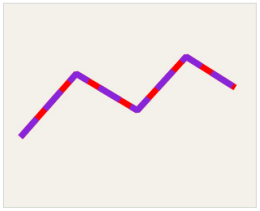

Native hedgerow with trees - associated with bank or ditch


Ecologically valuable line of trees

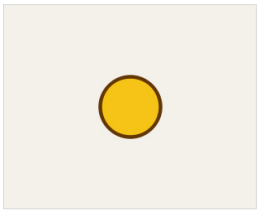

Ecologically valuable line of trees - associated with bank or ditch

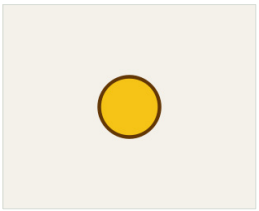

Line of trees

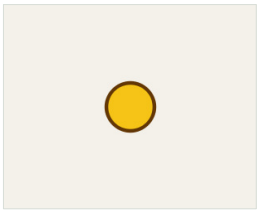

Line of trees - associated with bank or ditch

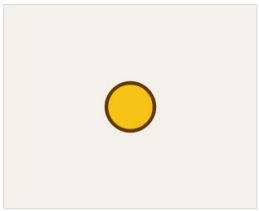

Non-native and ornamental hedgerow

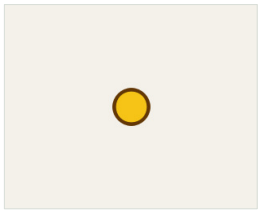
Individual Trees (marker size follows size class)

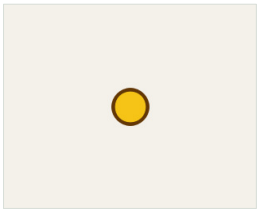

Very Large Urban Tree

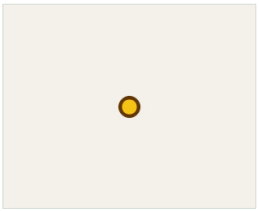

Very Large Rural Tree

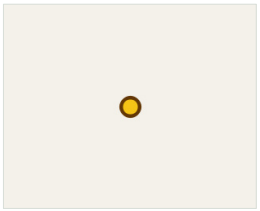

Large Urban Tree


Large Rural Tree


Medium Urban Tree


Medium Rural Tree


Small Urban Tree


Small Rural Tree

B Feature symbol library

The Feature Mapper's full point-symbol library (167 types), grouped as the picker groups them. Line and area feature types are listed after the symbols.

Species Records

Bats


PRF


Roost Entry / Exit Point


Confirmed Roost

Birds


Bird Nest (Active)


Bird Nest (Inactive)


Bird Pellets


Heronry


Schedule 1 Nest

Mammals


Badger


Deer


Fox


Hedgehog


Squirrel


Brown Hare


Pine Marten

Amphibians & Reptiles


Amphibian Eggs

Invertebrates


Stag Beetle


White-clawed Crayfish


Glow Worm

Reptiles


Slow-worm


Common Lizard


Grass Snake


Adder

Amphibians


Great Crested Newt


Common Frog


Common Toad


Smooth Newt


Palmate Newt

Field Signs

Mammal Signs


Badger Latrine


Deer Droppings


Fox Droppings


Otter Spraint


Otter Footprints


Water Vole Latrine


Water Vole Droppings


Water Vole Feeding Station


Hedgehog Droppings


Mink Scat

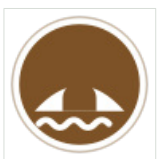

Pine Marten Scat


Mole Hill


Gnawed Hazelnut

Structures & Burrows

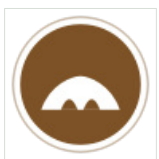

Badger Sett

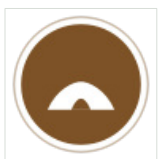

Otter Holt

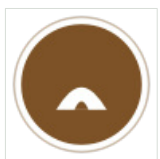

Water Vole Burrow


Rabbit Burrow


Main Sett


Annexe Sett


Subsidiary Sett


Outlier Sett


Otter Couch


Otter Slide


Water Vole Runs


Rabbit Warren


Brown Hare Form


Squirrel Drey


Harvest Mouse Nest


Dormouse Nest


Wood Ant Nest

Bat Signs


Bat Droppings


Bat Urine Staining


Bat Feeding Remains


Bat Corpse

Badger Signs


Badger Snuffle Hole


Badger Footprints


Badger Hair


Badger Push-through

Reptile & Amphibian Signs


Sloughed Skin

Invertebrate Signs


Brown Hairstreak Egg

Existing Features

Water


Water Body


GCN Pond


Kingfisher Bank

Vegetation & Deadwood


Veteran Tree


Tree


Tree Proposed for Removal


Tree Stump


Dead Wood


Log Pile


Brash Pile


Ancient Tree


Notable Tree


TPO Tree


Standing Deadwood / Snag

Amphibians & Reptiles


Herptile Refuge


Reptile Hibernaculum

Built


Green Roof

Botanical


Notable Plant


Ancient Woodland Indicator

Bat Survey

Equipment


Static Bat Detector


Handheld Bat Detector


Bat Box

Survey Positions


Camera


Thermal Camera


Surveyor


Observer Station


Suggested Surveyor Position

Building Roost Features


Gap Under Tiles


Gap Under Ridge Tiles


Slipped / Raised Tile


Gap Under Flashing


Lifted Roofing Felt


Gap Under Soffit


Gap Behind Fascia / Barge Board


Gap Under Weatherboarding


Missing / Eroded Mortar


Gap Under Coping Stones


Air Brick / Vent


Cavity Wall Access


Gap Around Window / Door Frame


Loft Access Point

Tree Roost Features


Tree Cavity / Rot Hole


Tree Split / Crack


Lifting Bark


Woodpecker Hole


Dense Ivy

Enhancements

Bats


Integrated Bat Tube


Bat Access Tile

Birds


Bird Box


Swift Brick


Swift Box


Barn Owl Box


House Martin Cup


Sparrow Terrace


Starling Box


Kestrel Box


Tawny Owl Box


Little Owl Box


Sand Martin Wall


Tern Raft

Invertebrates


Bee Brick


Bee Bank


Insect Box


Stag Beetle Loggery

Mammals


Wildlife Gap


Hedgehog House


Hedgehog Gap


Dormouse Box


Badger Gate


Artificial Sett


Dormouse Bridge


Pine Marten Den Box

Amphibians & Reptiles


Grass Snake Egg-laying Heap


Gully-pot Ladder

Survey & Design

Site


Building


Target Note

Survey Positions

Survey Methods

Reptiles


Reptile Refugium / Mat

Amphibians / GCN


eDNA Sample Point


Bottle Trap


Torch Count Point


Egg Search Point


Pitfall Trap

Dormouse


Dormouse Nest Tube


Dormouse Footprint Tunnel

Mammals


Mink Raft


Hedgehog Footprint Tunnel

Botanical



Quadrat

General



Photo Viewpoint



Sample Point

Invasive Species

Terrestrial Plants



Japanese Knotweed



Himalayan Balsam



Giant Hogweed



Rhododendron



Cotoneaster



Buddleia



Montbretia

Aquatic Plants



New Zealand Pigmyweed



Floating Pennywort

Animals



American Mink

Line feature types

Type	Style
Amphibian Exclusion Fencing	#aa2200, dash dot
Animal Worn Path	#8B5e3c, dash

Bat Commuting Route	#2233aa, dash
Bat Flight Line	#4455cc, dot
Connectivity Corridor	#2e8b2e, solid
Gap Along Ridge	#8A7ED8, dash
Gap Under Eaves	#5B4FA0, dash
Green Corridor	#66bb44, solid
Habitat Management Zone	#4a8c2a, dash
Hanging Tiles	#F2C200, dash
Important Hedgerow	#2e7d32, solid
Line of Trees for Removal	#B3372F, dash
Line of Trees for Trimming / Pruning	#C97B16, dash
Newt Tunnel	#228b44, dash dot
Otter Ledge	#3F8266, dash
Reptile Exclusion Fencing	#cc4400, dash dot
Riparian Buffer	#3F8266, solid
SuDS / Drainage Feature	#0077aa, solid
Survey Transect	#8A968D, dot
Temporary Fencing	#C97B16, dash

Area feature types

Type	Fill / border
Amphibian Receptor Site	#44bb88 / #228855, dash
Badger Territory	#8B6340 / #5a3e10, dash
Bat Activity Survey Zone	#4455cc / #222288, dot
Bat Foraging Area	#221166 / #110044, dash
Biosolar Roof Zone	#ffdd44 / #cc9900, solid
Camera Field of View	#2a6fd0 / #103a78, solid

Cellar / Basement Void	#7A6FC4 / #4A3F94, dash
Construction Exclusion Zone	#ff4400 / #cc2200, dash
Dormouse Home Range	#d4aa40 / #8B6320, dash
Dusk / Dawn Survey Zone	#332266 / #221144, dot
Ecological Mitigation Zone	#ffcc00 / #cc9900, dash dot
Flood Risk Zone	#aaddff / #4499cc, dash
GCN Terrestrial Zone	#44bb44 / #228822, dash dot
Green Roof Footprint	#44bb88 / #228855, solid
Hibernaculum Creation Zone	#cc8844 / #886622, dash dot
Invasive Species Stand	#cc5555 / #992222, dash dot
Japanese Knotweed Stand	#cc3333 / #991111, dash dot
Lake	#1a60a0 / #0e3a6a, solid
Living Wall Area	#66dd66 / #33aa33, solid
Loft Space	#41D6E8 / #00B8CE, solid
Native Planting Zone	#44aa22 / #226611, dash
Observer Field of View	#888844 / #555522, dot
Orchard / Fruit Tree Area	#88cc44 / #448822, dash
Otter Spraint Territory	#3F8266 / #1F4A37, dash
Phase 2 Survey Area	#aaaaaa / #666666, dash dot
Pond	#2a7fc4 / #1F4A37, solid
Protected Area	#44ccaa / #228866, solid
Protected Habitat	#22aa44 / #116622, solid
Raptor Disturbance Buffer	#cc44aa / #881188, dash
Reptile Receptor Site	#44aa44 / #227722, dash
Rewilding / No-Mow Zone	#66bb44 / #338822, dash
Riparian Management Strip	#3F8266 / #1F4A37, solid
Roof Void	#41D6E8 / #0090A8, dash

Root Protection Area	#C97B16 / #cc5500, dash
SuDS Catchment Area	#0077aa / #004466, solid
Surveyor Field of View	#f2c200 / #9a7a00, solid
Thermal Camera Field of View	#d0562a / #7a2810, solid
Wader Scrape	#aaddff / #4499cc, solid
Walkover Survey Area	#8A968D / #555555, dot
Waterway	#4aabdd / #2a7faa, solid
Wildflower Seed Mix Area	#eebb22 / #aa7700, dash